Polycrystalline Sample Preparations Methods

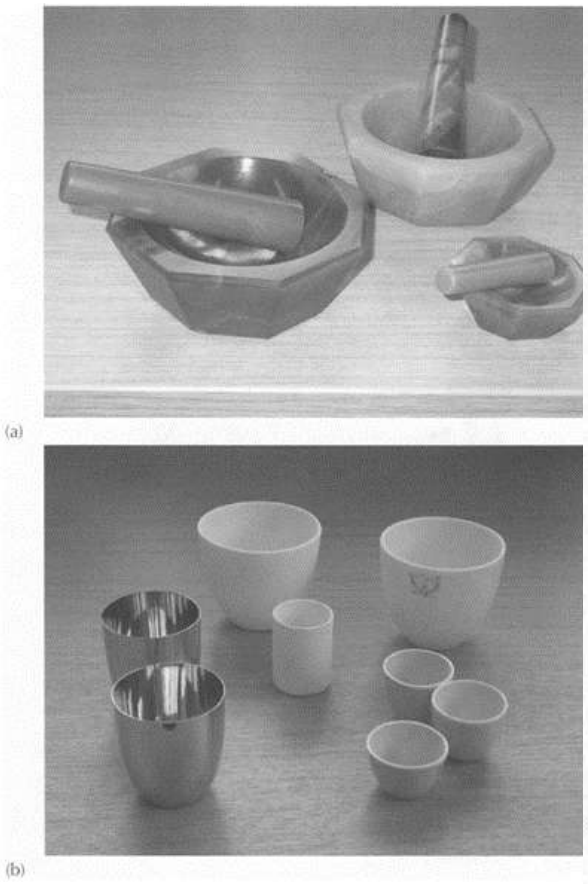
Solid State Reaction Method

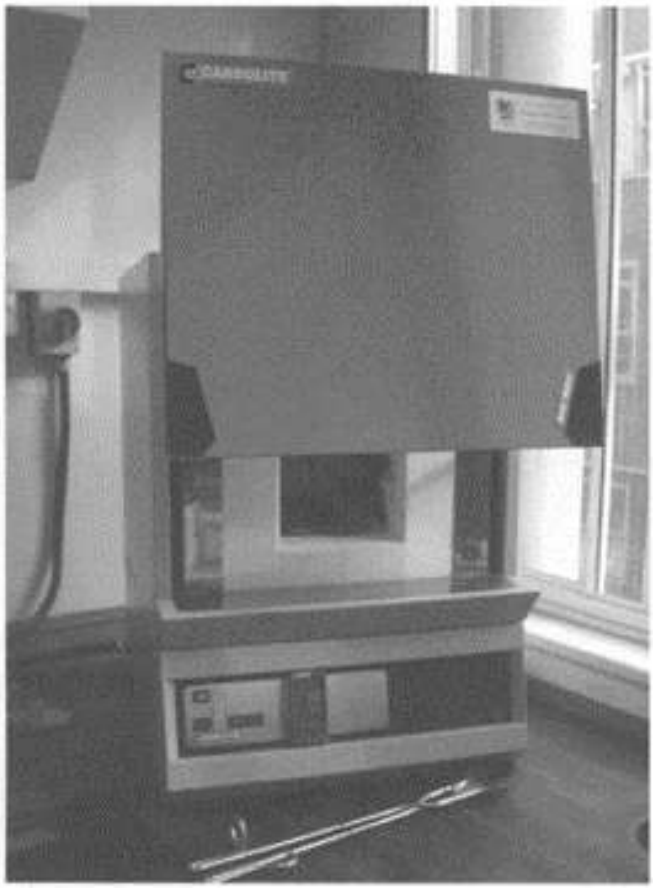
The procedure is to take stoichiometric amounts of the binary oxides, grind them in a pestle and mortar to give a uniform small particle size, and then heat in a furnace for several hours in an alumina crucible

Calcinations

Sintering

Reducing the particle size and lowering the temperature





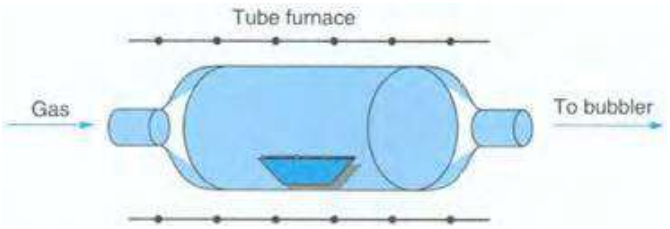
Sealed Tube Methods

Evacuated tubes are used when the products or reactants are sensitive to air or water or are volatile. An example of the use of this method is the preparation of samarium sulfide.

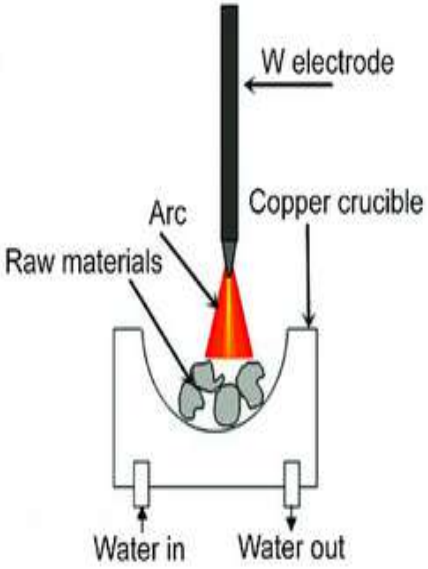
In this case, sulfur has a low boiling temperature (717 K) and an evacuated tube is necessary to prevent it boiling off and being lost from the reaction vessel.

Special Atmospheres

The preparation of some compounds must be carried out under a special atmosphere: an inert gas such as argon may be used to prevent oxidation to a higher oxidation state, an atmosphere of oxygen can be used to encourage the formation of high oxidation states; or conversely a hydrogen atmosphere can be used to produce a low oxidation state.



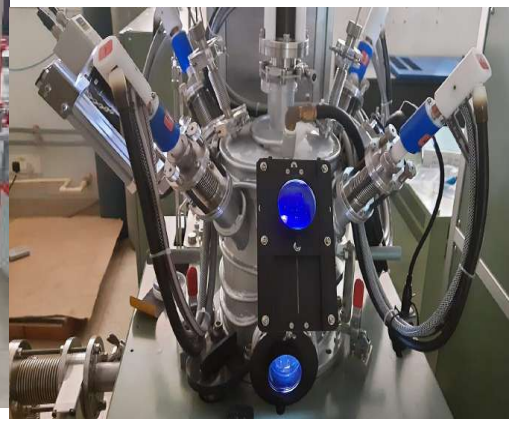
Arc Melting



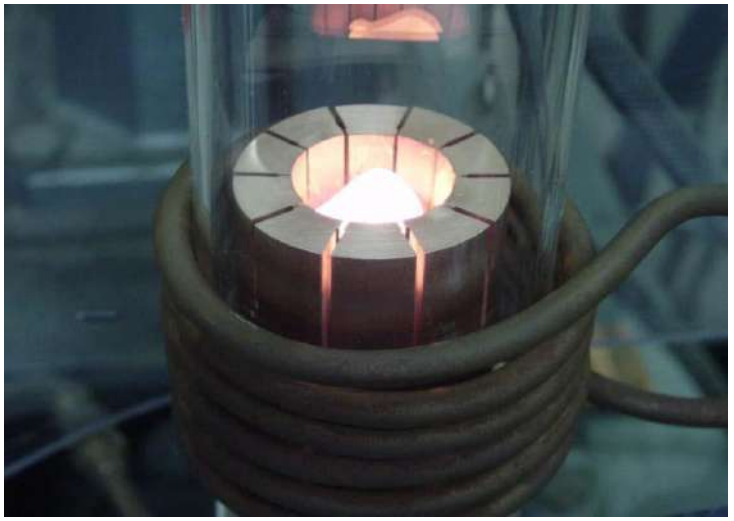


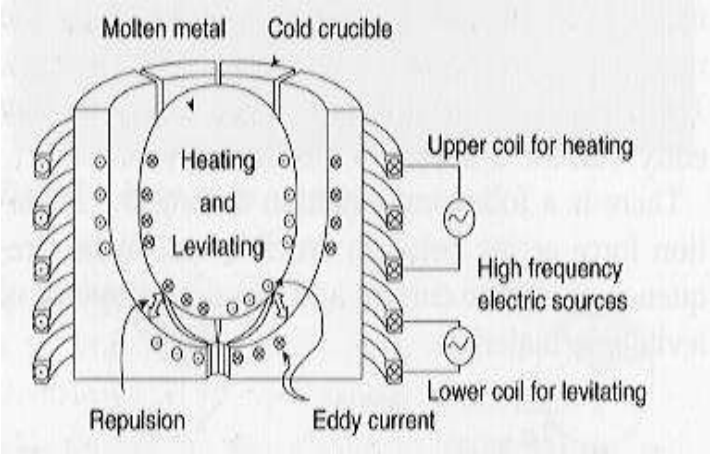






Induction Heating

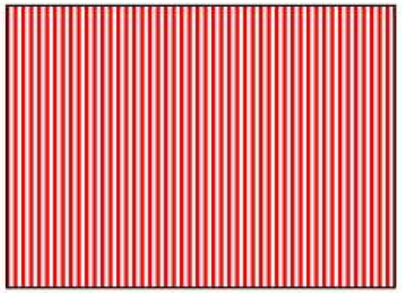




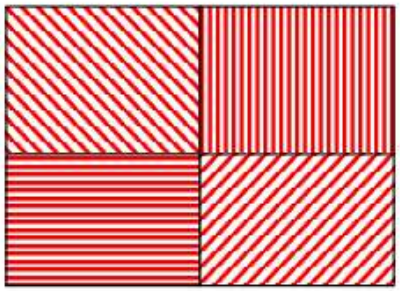




Why single crystals?



Single crystal



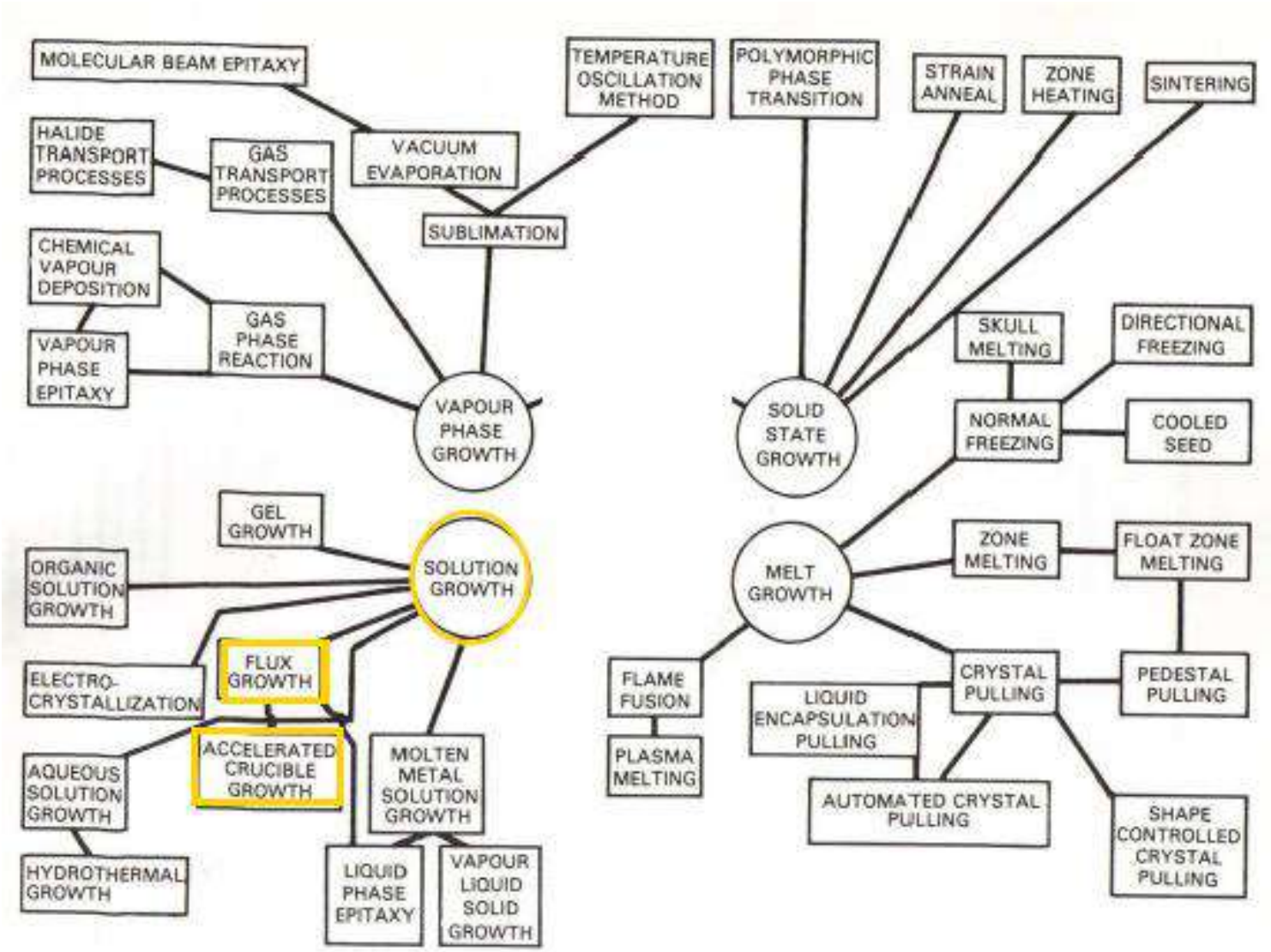
Polycrystalline

1. Anisotropic properties
2. No grain boundaries
3. Purity

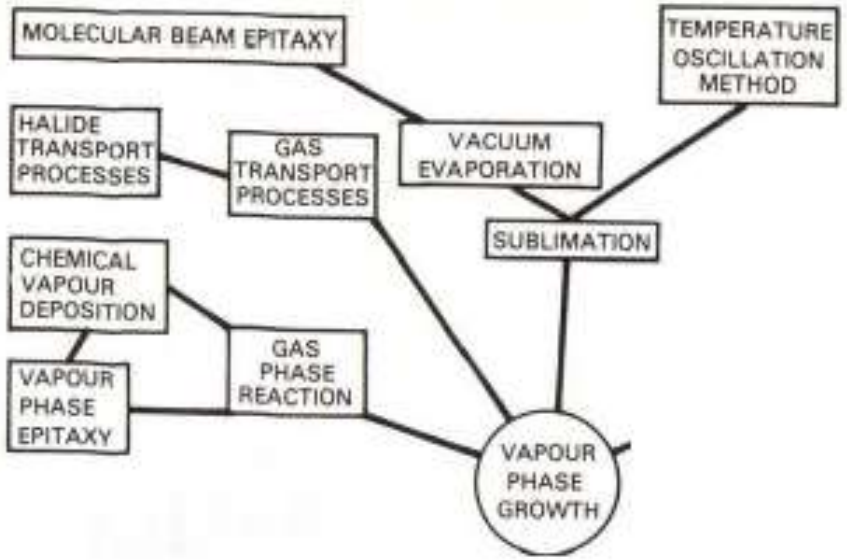
i.e. to measure the intrinsic physical properties you must use single-crystals

Single Crystal Growth

Crystal Growth Techniques



Crystal Growth Techniques



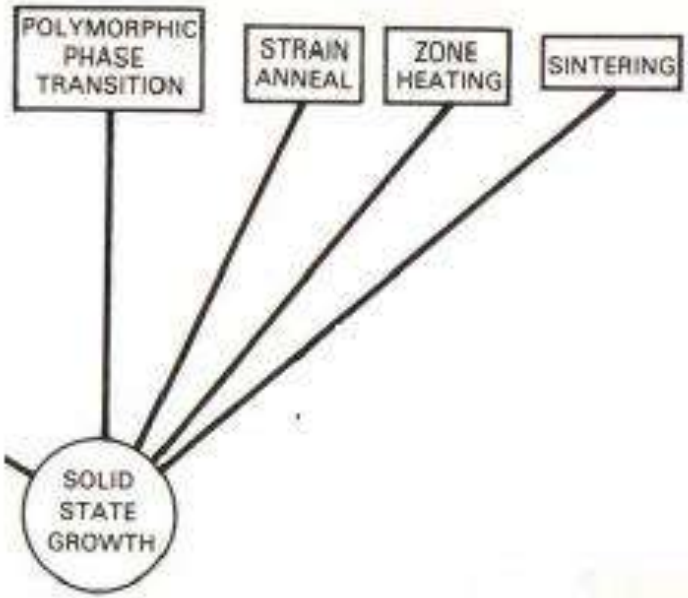
Growth from gas phase – 'precipitation' from gas phase

- solid → gas → xtal
- Chemical vapor transport – need a transport agent
- Pulsed laser deposition – need a substrate or seed

Issues:

- Induce a phase transition – homogeneous gas phase to solid + unsaturated gas
- Control conditions for supersaturation, usually by control of temperature gradients

Crystal Growth Techniques



Growth from a mixture of solids *(no melting)*

- kinetic and thermodynamic factors are important

Issues:

- Crystals are micron size
- Phase segregation, grain boundaries

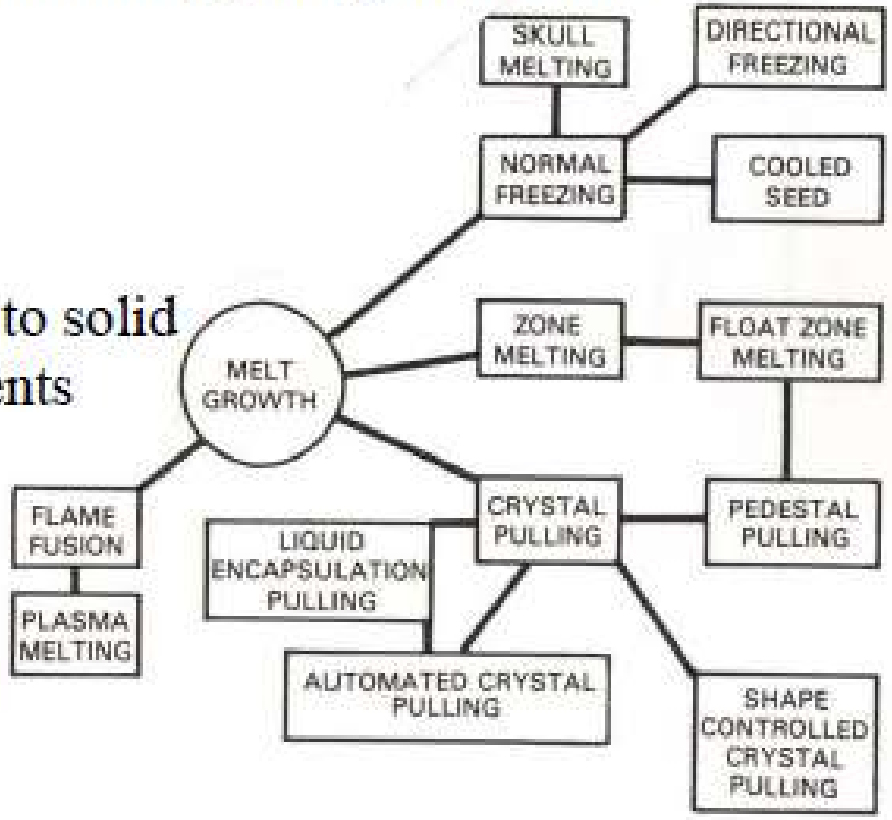
Crystal Growth Techniques

Growth from the liquid phase

- solid → liquid → solid
- Variations are Bridgman, Czochralski (pulling), Kyropoulos (top seeding), Verneuil (flame fusion), tri-arc, skull melting, image float-zone

Issues:

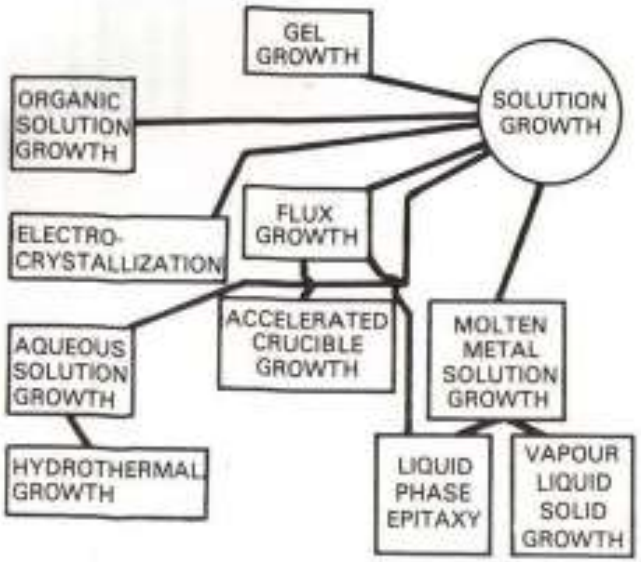
- Induce a phase transition – liquid to solid
- Control of temperature and gradients
- High melting points > 2000 °C
  (confirmation of melting!)
- Reactivity of crucible
- Control nucleation (seed crystal)



Crystal Growth Techniques

Growth from solution – precipitation from supersaturated solution

- Aqueous & organic solvents
- Inorganic solvent (flux, high T)
- Hydrothermal (high P & T)



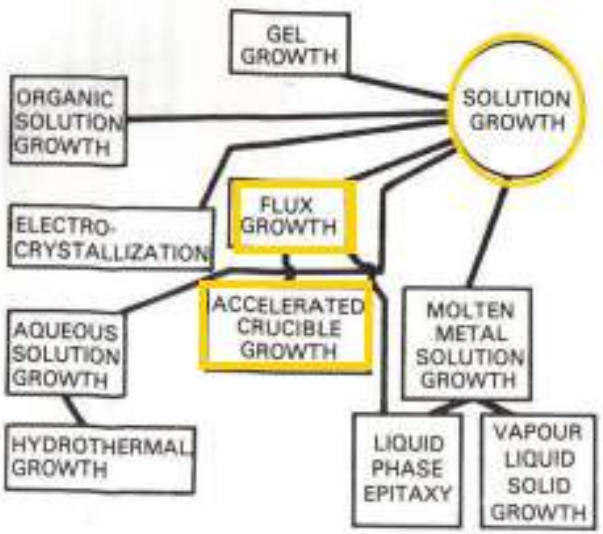
Issues:

- induce a phase transition – homogeneous solution to solid + unsaturated solution
- control conditions for supersaturation– concentration (evaporation of solvent), temperature (solubility or saturation is a function of temperature)
- reactivity of container (crucible)
- inclusion of solvent in crystals

Crystal Growth Techniques

Advantages:

- Grow congruently and incongruently melting materials
- Need relatively simple equipment
- Has short growth-time scales
- Need small amounts of materials



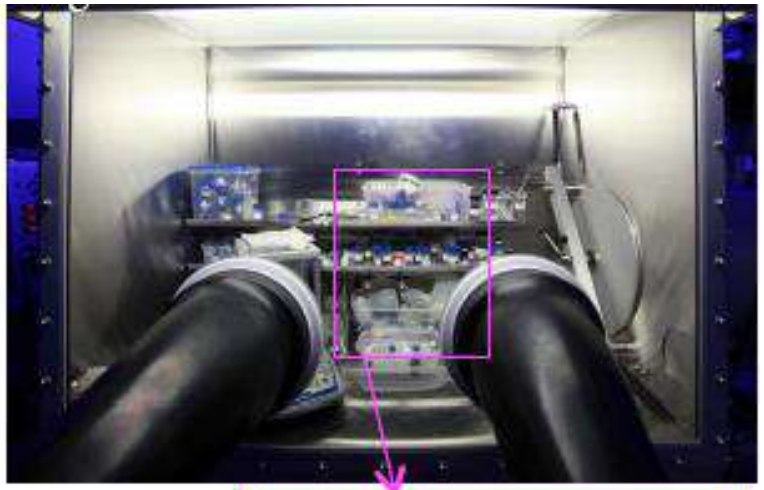
Disadvantages:

- Not too large a crystal (mm to cm)

Flux Method

- Elements
- Cutting tools



















- Crucibles
- Tubes (reaction under ambient conditions not possible)

	T_{max} (°C)	$T_{melting}$ (°C)
borosilicate glass (Pyrex)	515	820
gold	1013	1064
silica (quartz)	1200	1853
platinum	1720	1770
alumina (Al_2O_3)	1900	2072
zirconia (ZrO_2)	2000	2700
magnesia (MgO)	2400	2852
tantalum	1400	3017





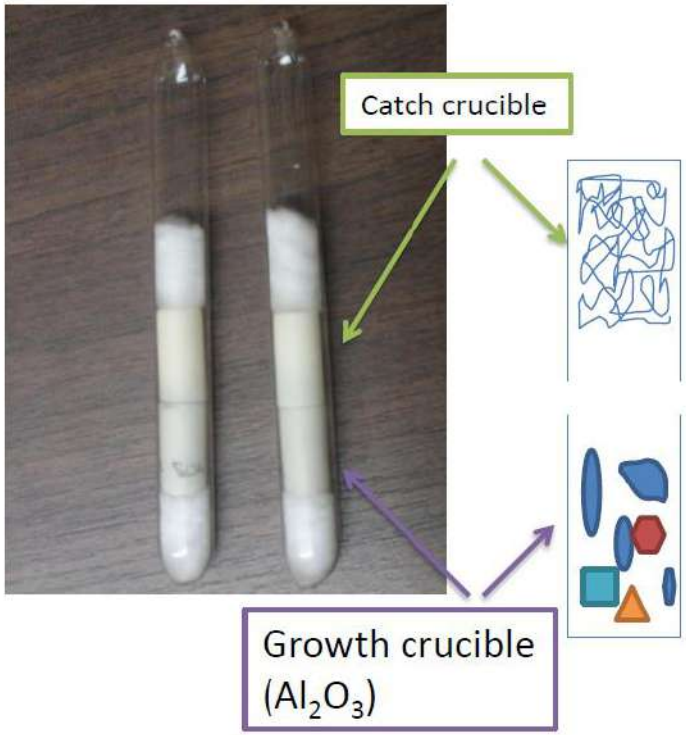


- Crucibles
- Tubes (reaction under ambient conditions not possible)

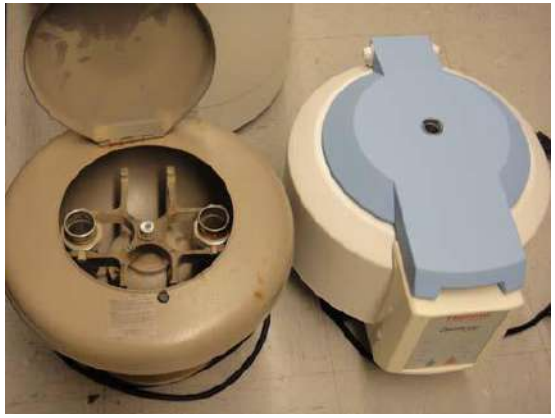
Elements	Container & tube choices
Alkali & alkaline-earth metals	Ta, steel
Al, Ga	Al_2O_3, MgO, BeO
Mg	MgO, Ta, graphite or steel
Cu, Ag, Au	graphite, MgO, Al_2O_3, Ta
Fe, Co, Ni	Al_2O_3, ZrO_2
Zn, Cd, Hg	Al_2O_3
In	Al_2O_3, Ta
Rare-earth metals	Ta, Mo, W, BeO
Bi, Sn	Al_2O_3, SiO_2, graphite
Sb	SiO_2, graphite



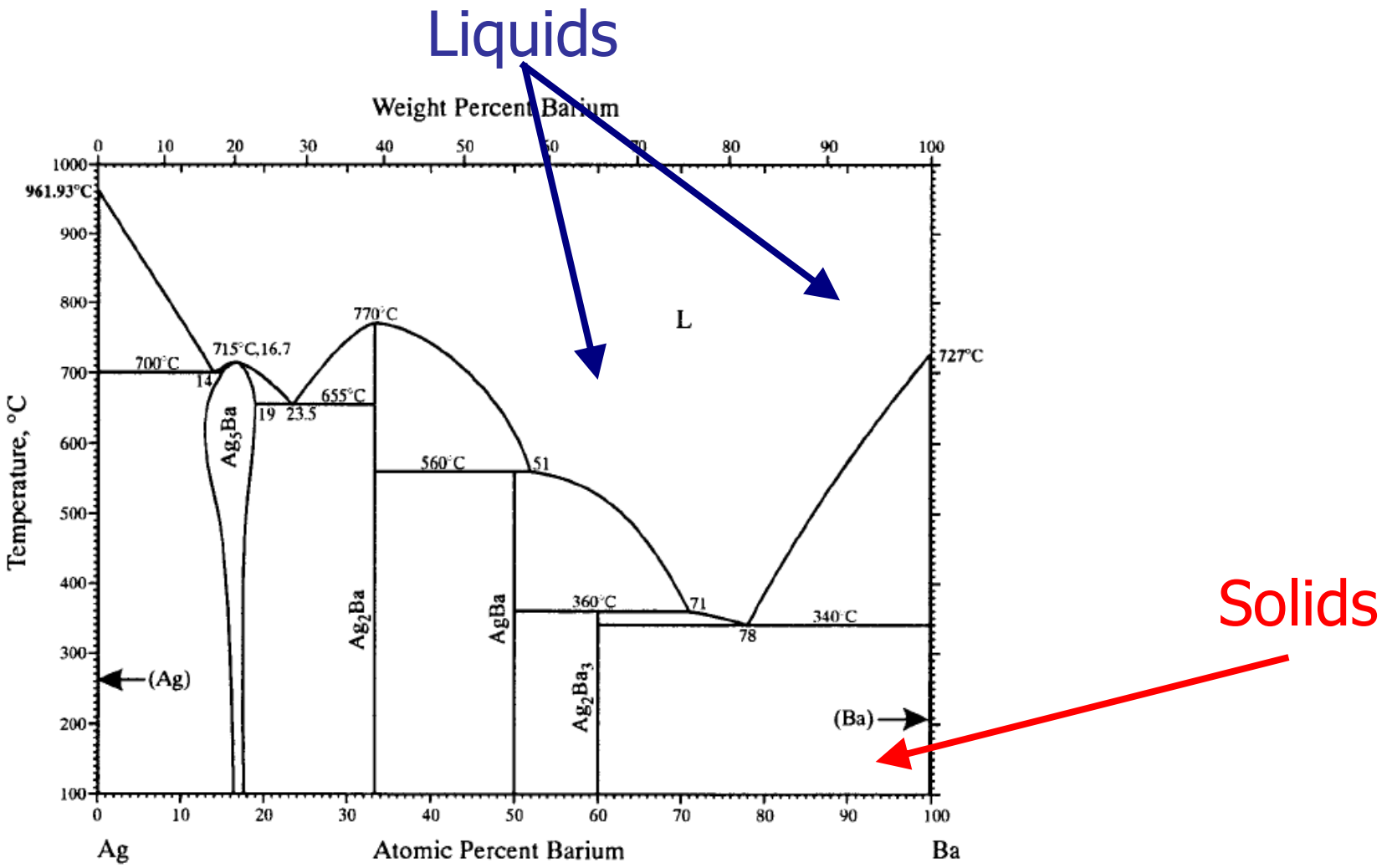



Catch crucible
Growth crucible
(Al_2O_3)

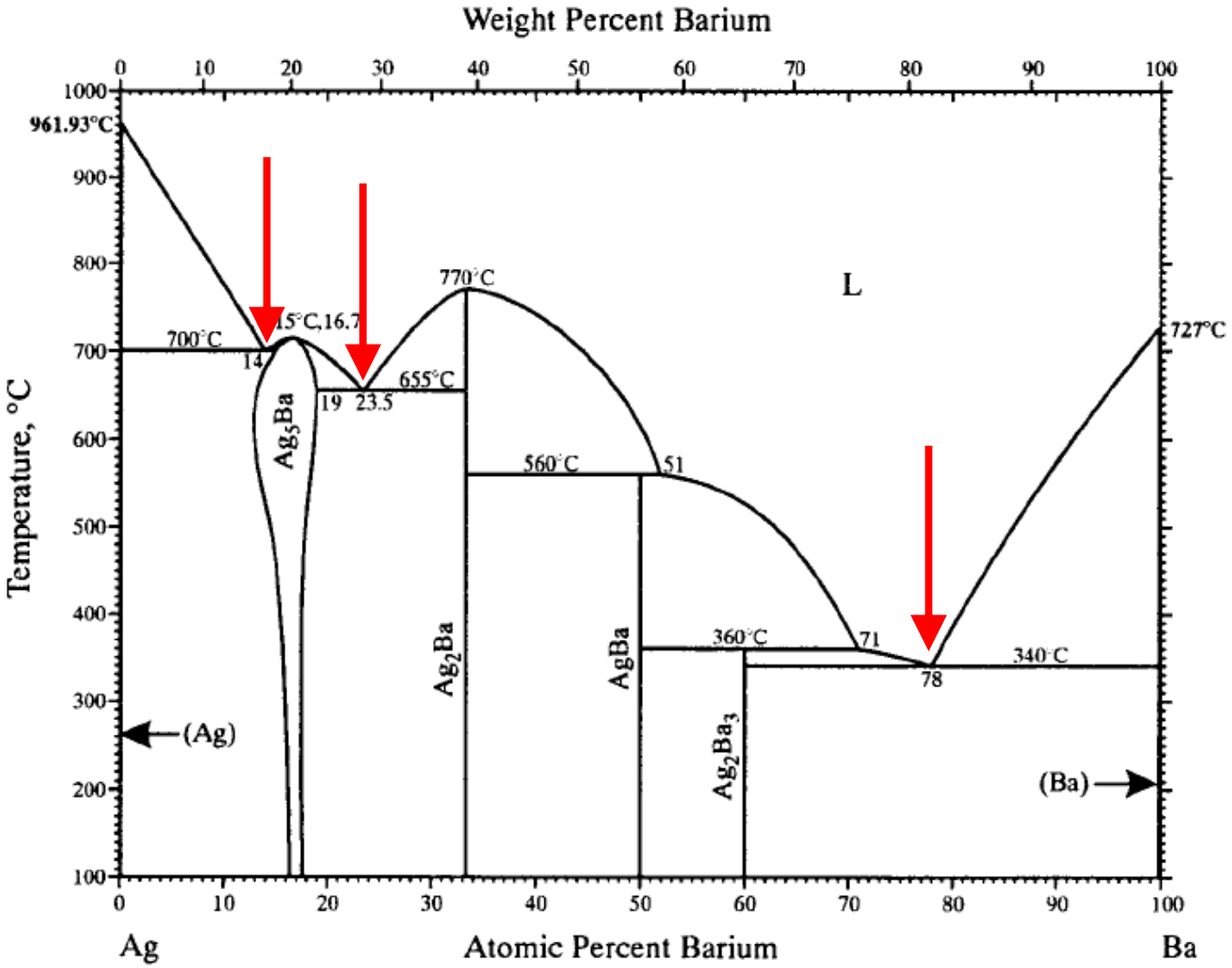

CARBOLITE



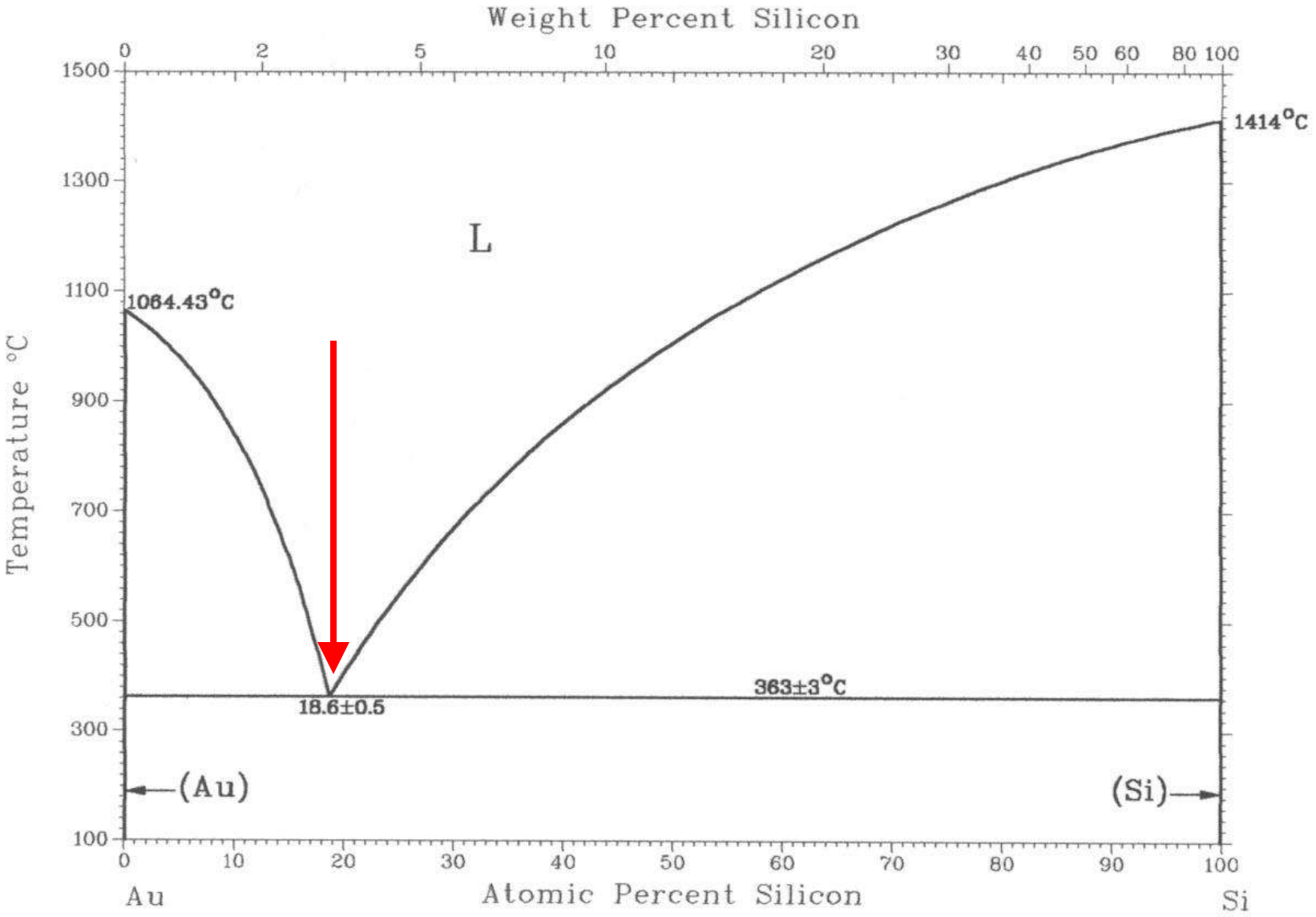
Binary phase diagrams are fundamentally like maps: they show the extent of liquid and solid.



Eutectic points: minima in liquid regions

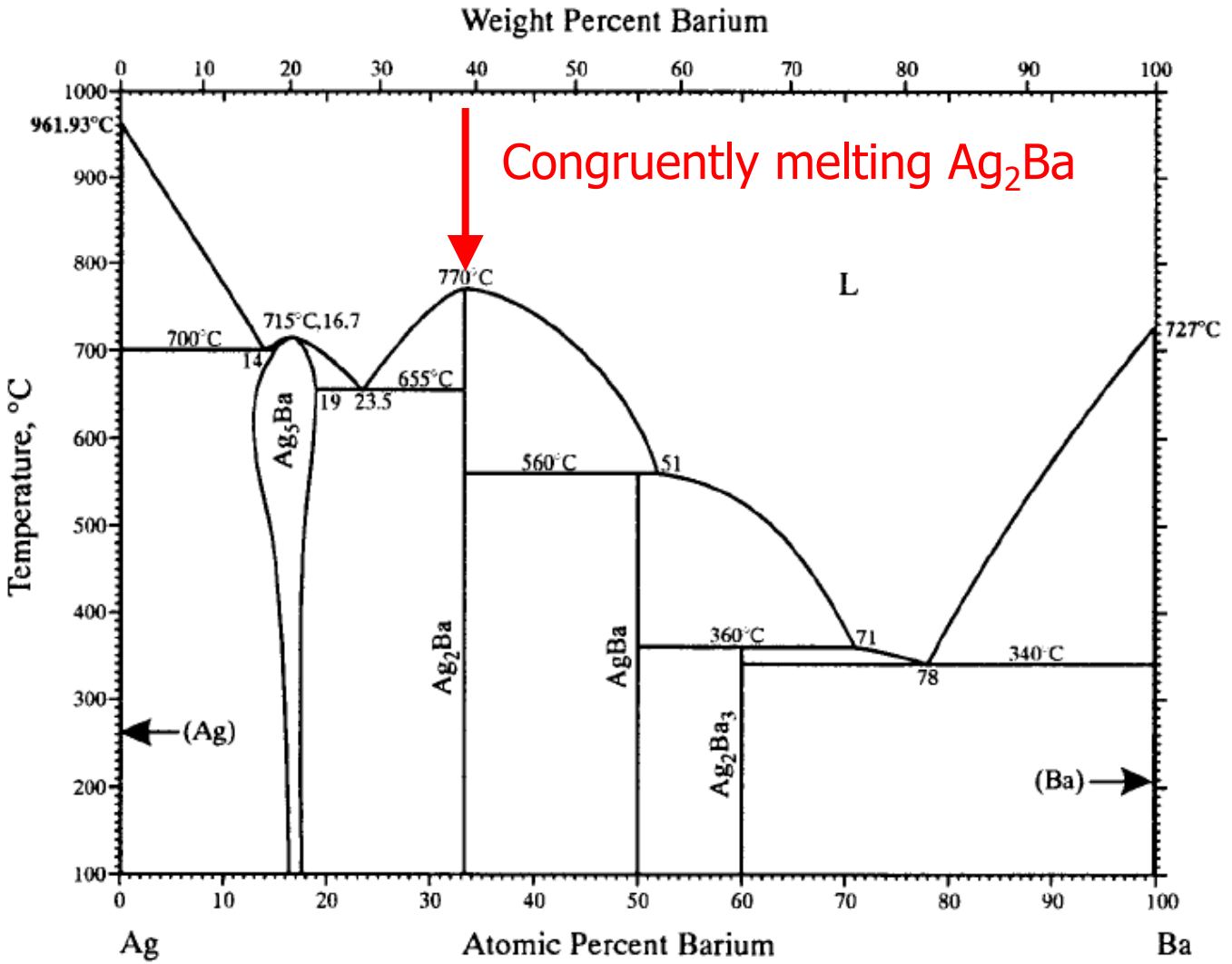


Deep Eutectics offer tempting liquid regions for growth



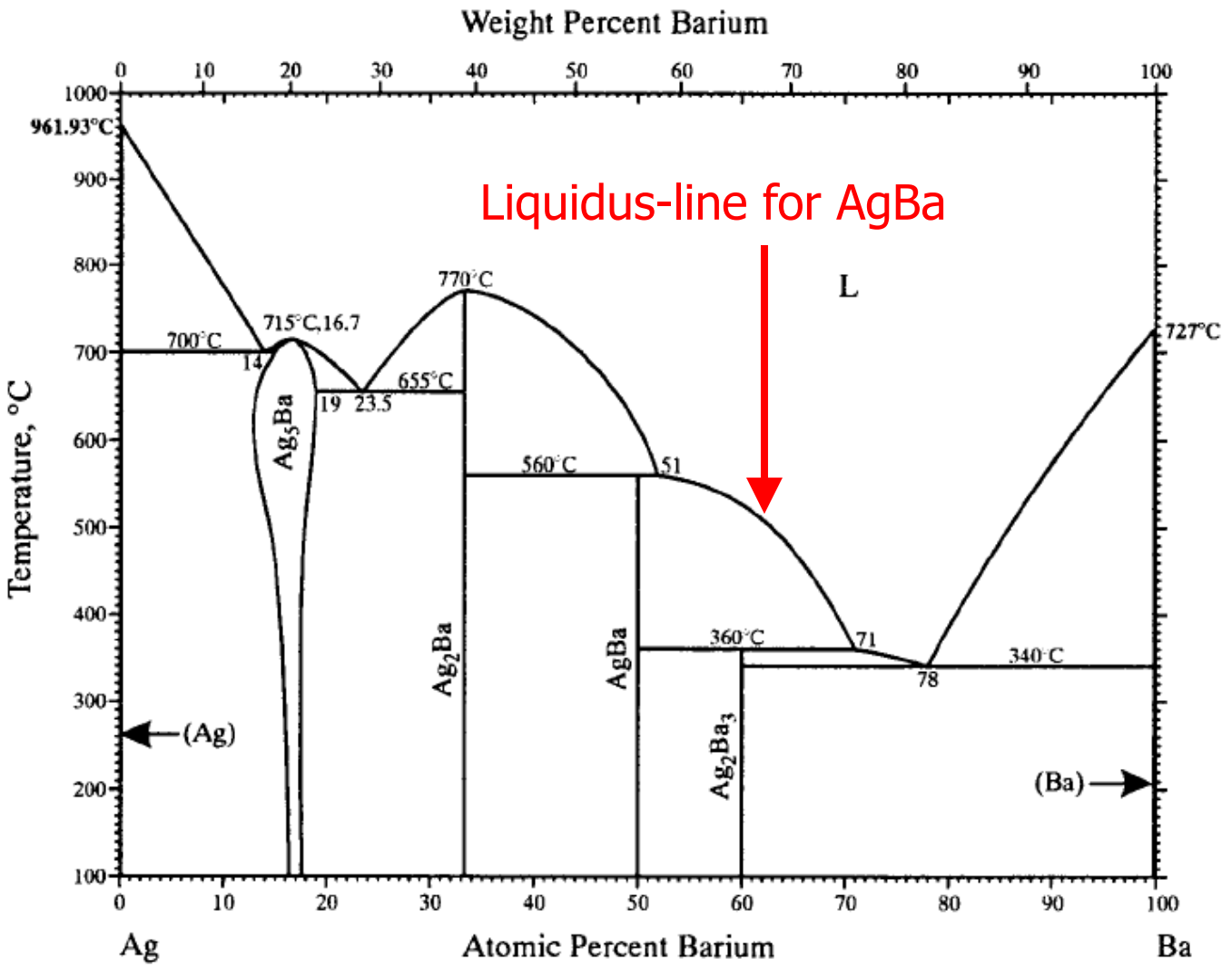
Congruently melting compounds transform from a homogeneous solid to a homogeneous liquid at melting point.

Congruently melting compounds can be made by a wide variety of growth techniques that simply melt and solidify samples of fixed composition.



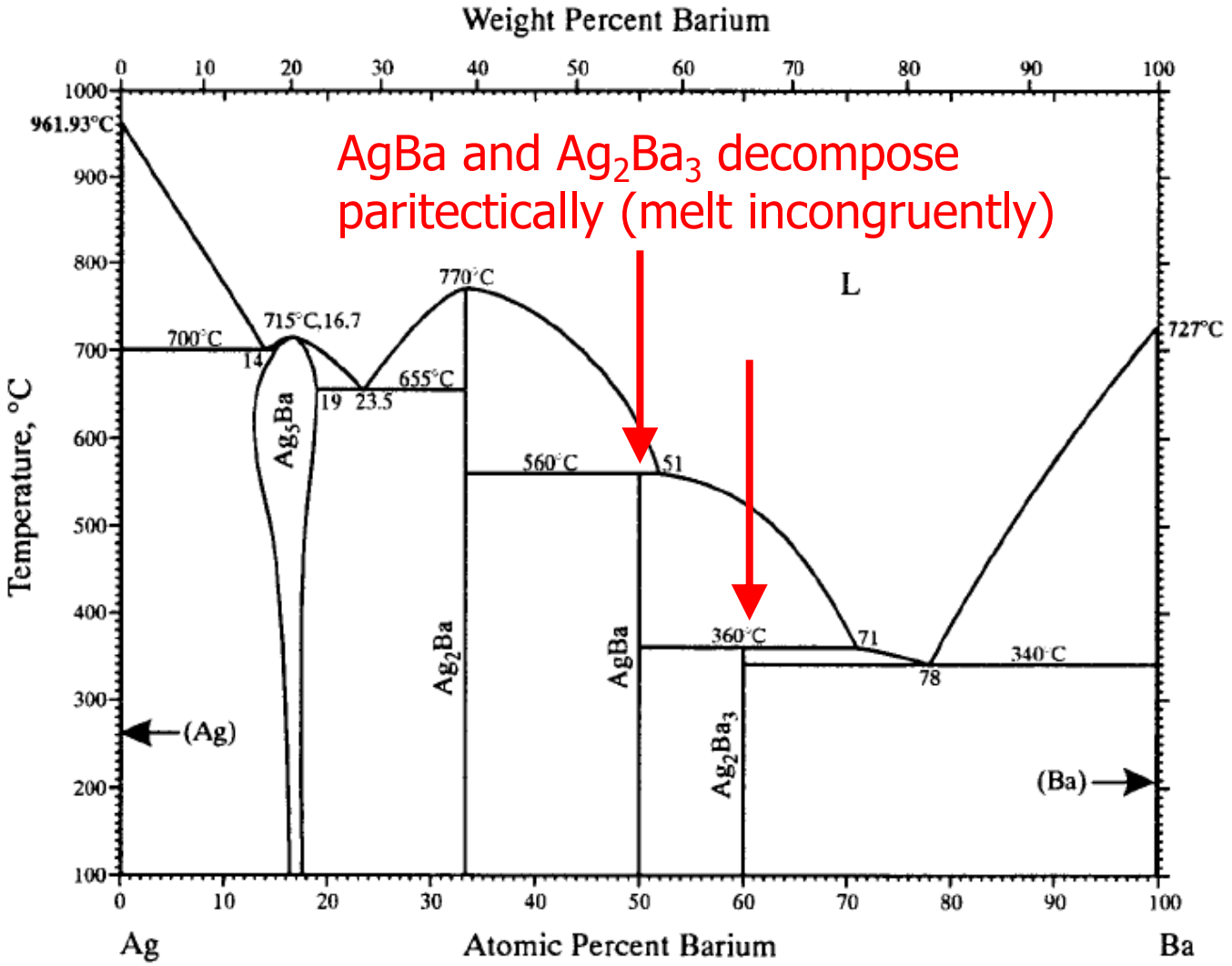
Liquidus-line

As we cool through the liquid, we ultimately cool enough to hit the liquidus-line for AgBa. At this temperature AgBa starts crystallizing and the remaining liquid becomes more Ba rich. The sample is no longer homogeneous and instead contains a solid of one stoichiometry and a liquid of another, changing stoichiometry.



Incongruently melting compounds undergo a paritectic decomposition into a mixed solid and liquid phase, only to form a homogeneous liquid at higher temperatures.

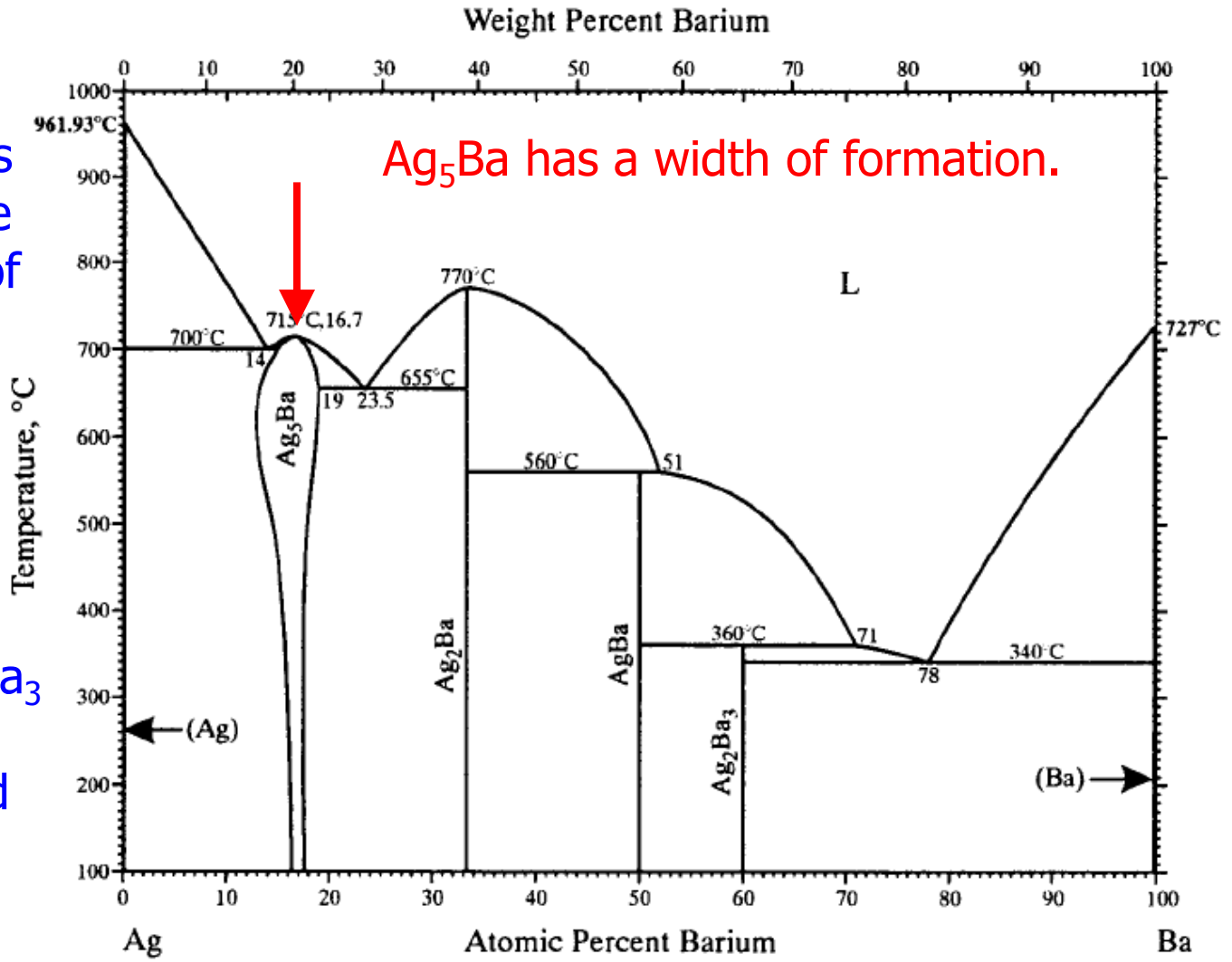
If you cool a liquid with composition AgBa, it will first form Ag_2Ba as the remaining liquid becomes more Ba rich and only forms AgBa below the paritectic temperature of 560 C.



Widths of formation and line compounds.

Ag_5Ba can form with a variety of compositions and even have a single crystal with a spread of stoichiometries.

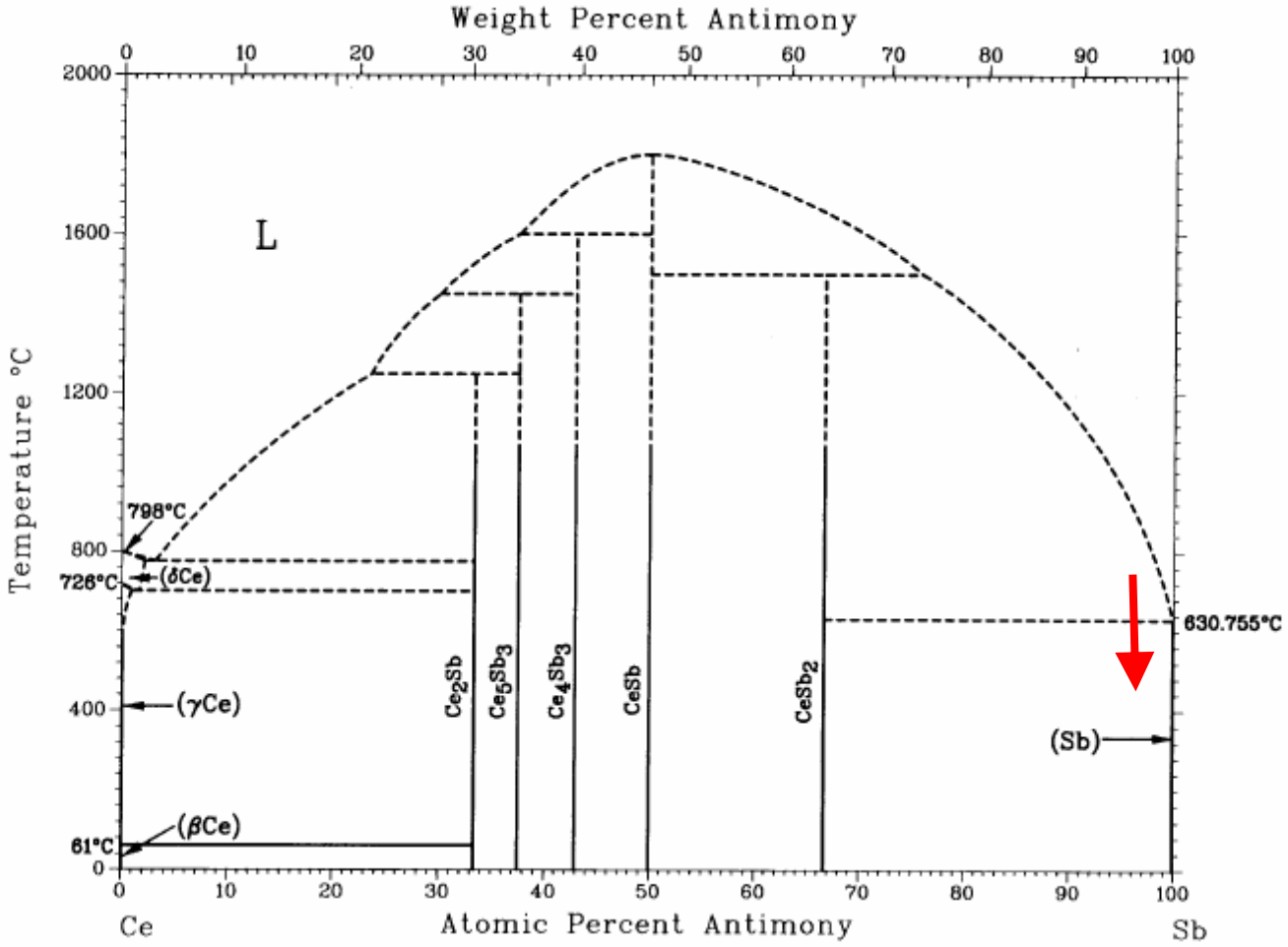
Ag_2Ba, AgBa and Ag_2Ba_3 are show to have no width of formation and are called "line" compounds.



Crystal growth of $CeSb_2$?

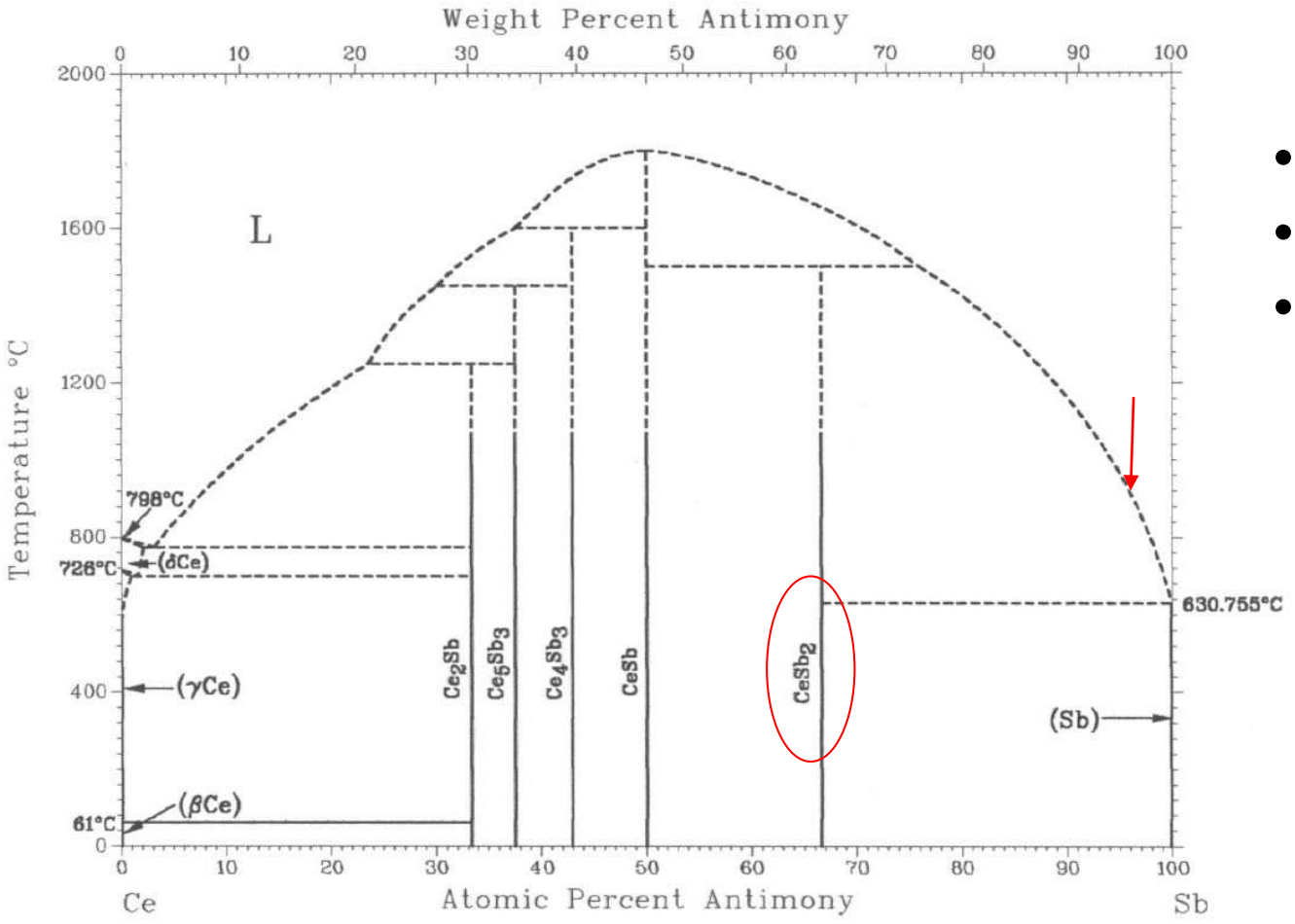
$CeSb_2$ is incongruently melting at a relatively high temperature. An attempt to cool a melt of $CeSb_2$ would end up with a mixed phase and lots of mess (high vapor pressures).

On the other hand there is a very open line of primary solidification. Grow $CeSb_2$ out of a "self flux" of excess Sb.



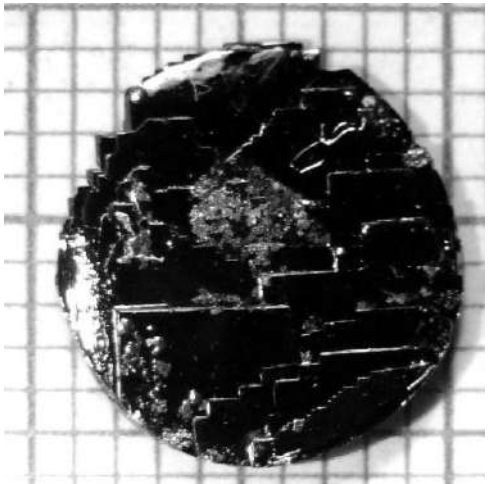
Flux growth (slow cooling of a melt)

Basic idea: slow cool into 2-phase region



eg: $CeSb_2$ / Sb

- self flux
- $Ce_{0.05}Sb_{0.95}$
- 1190 °C → 700 °C

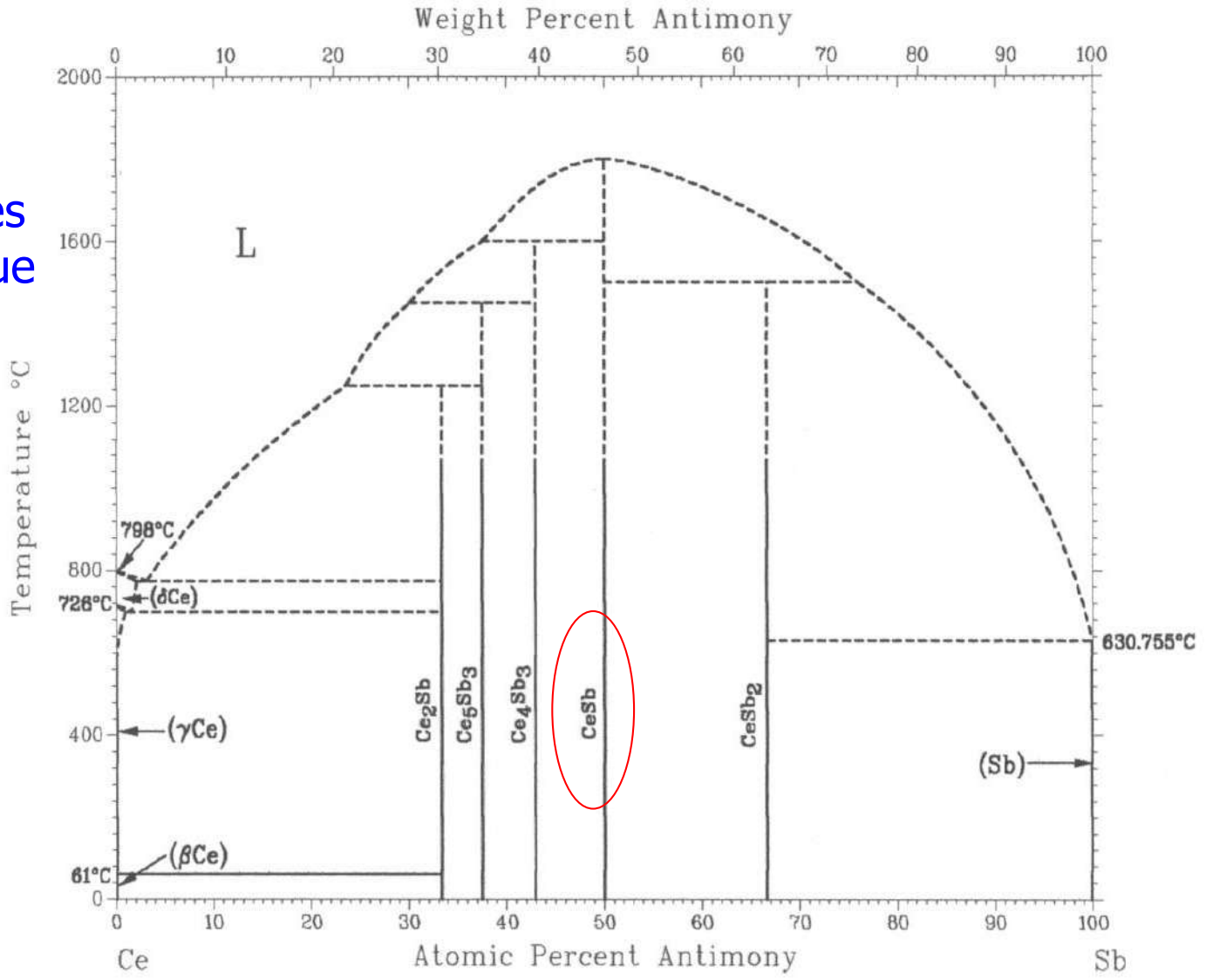


Growth of the congruently melting compound, CeSb?

This is tricky if done just out of the binary: very high temperatures and lots of defects (due to vapor pressure and entropy).

Can this be grown out of extra elements in manner similar to growing a salt out of water?

This question is the essence of flux growth.



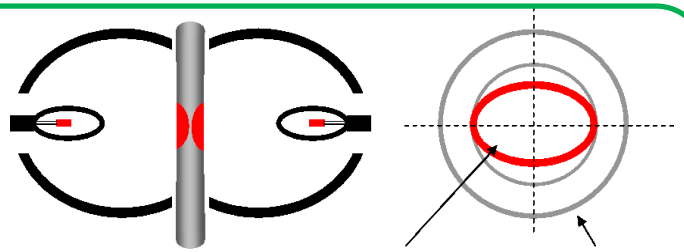
Floating Zone Method

Single Crystal Growth Using Floating Zone Furnace

Technique ideal for producing large volume, high quality crystals

No crucibles and no contamination with flux adhering to crystal surfaces

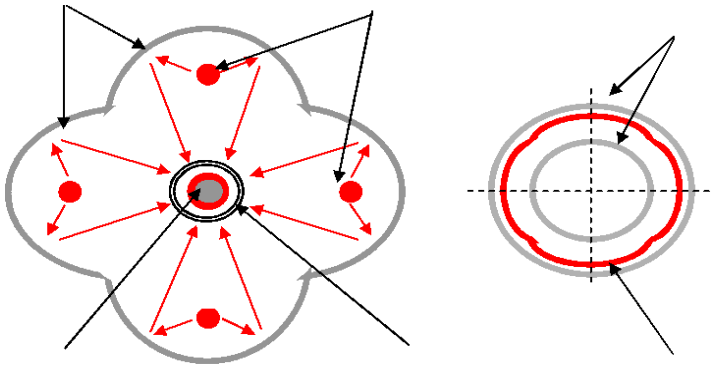
2 Mirror Halogen Lamp Furnace

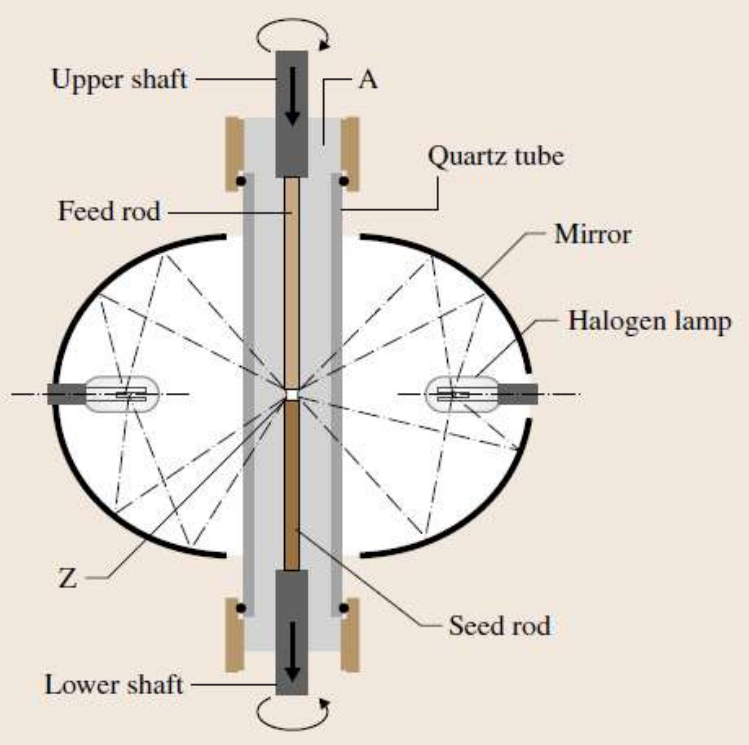


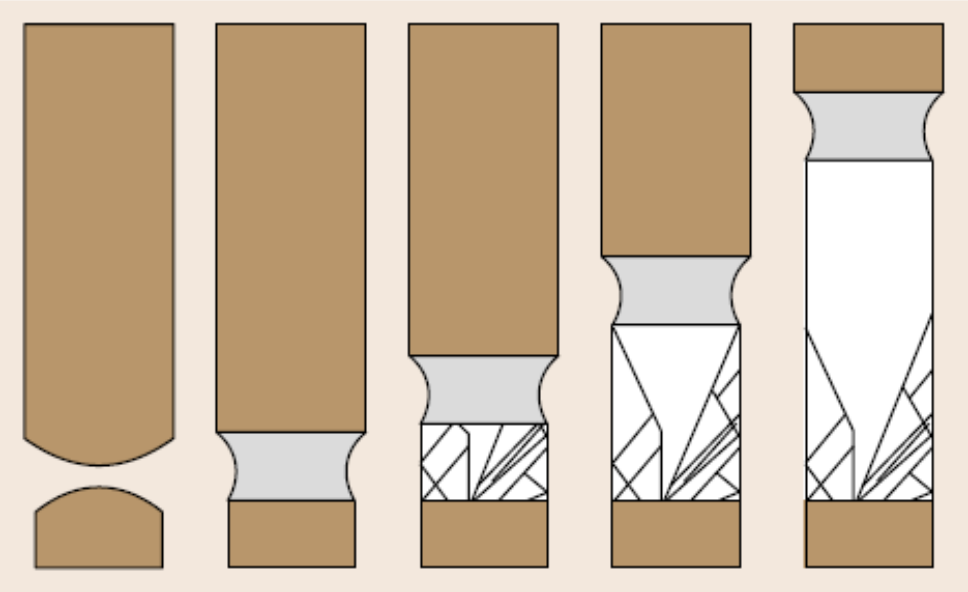
Temperature Profile in not very uniform in 2 Mirror Furnace

4 Mirror Halogen Lamp Furnace

4 Mirror Xenon Lamp Furnace

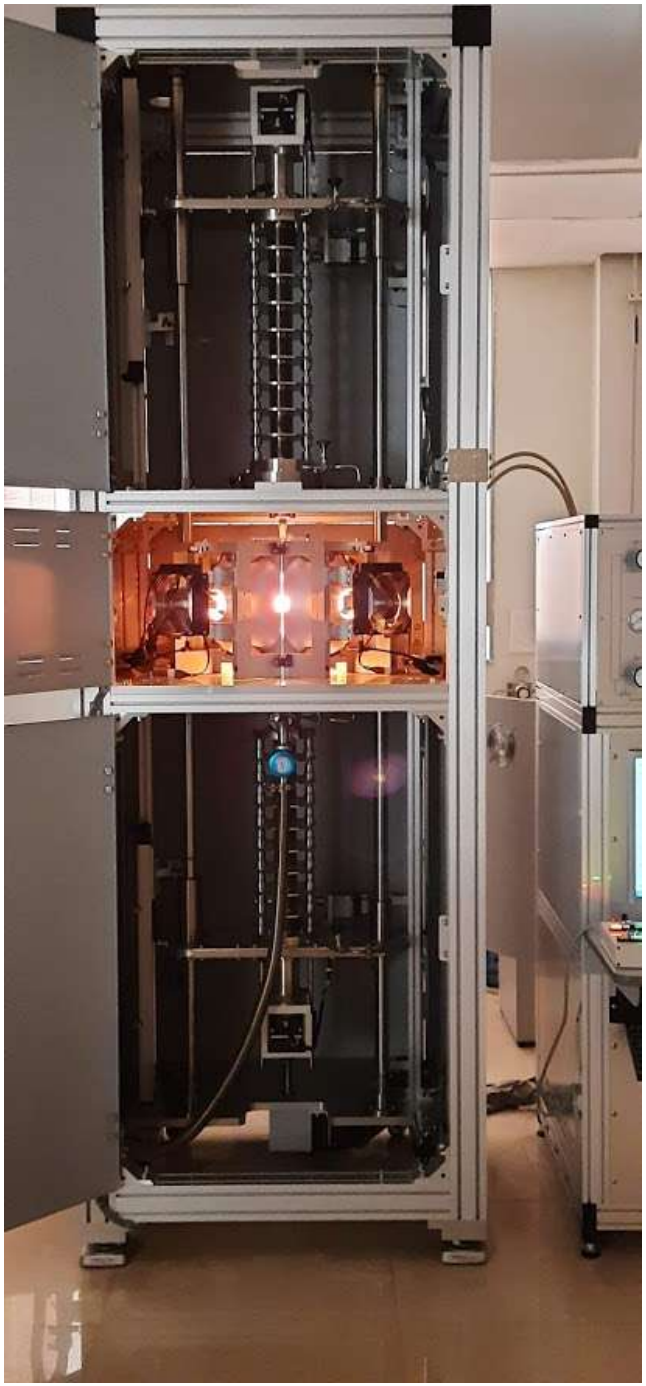






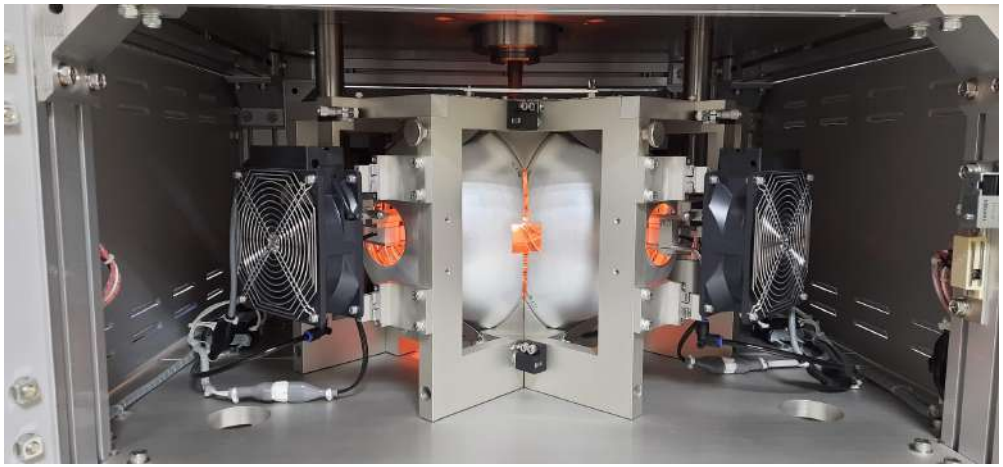
http://www.youtube.com/watch?v=L8XTTObVNZ8

http://www.youtube.com/watch?v=XbkSaEk2lLo

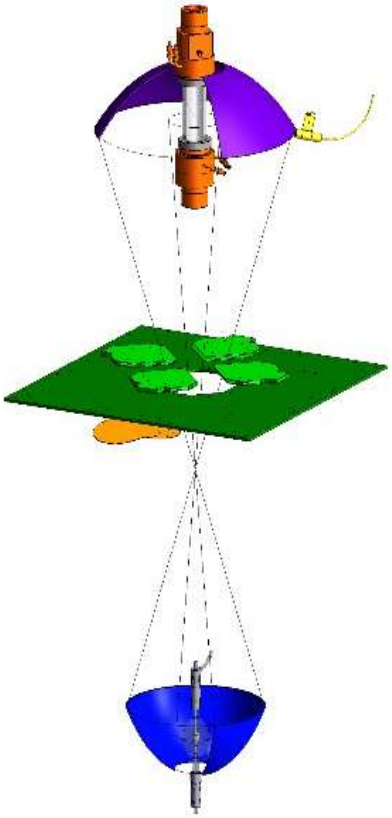


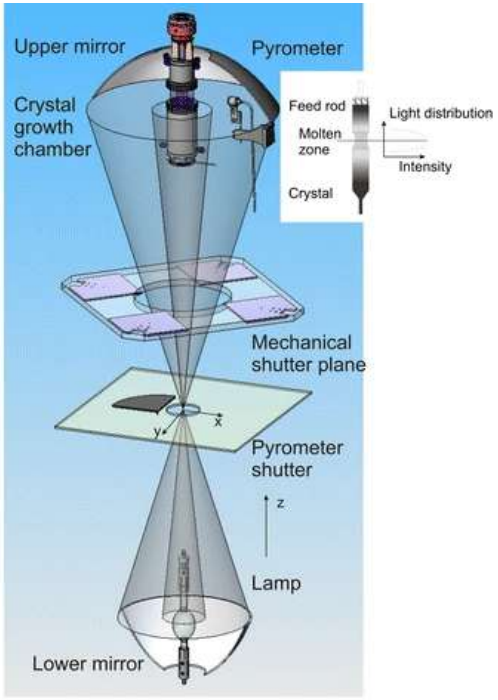


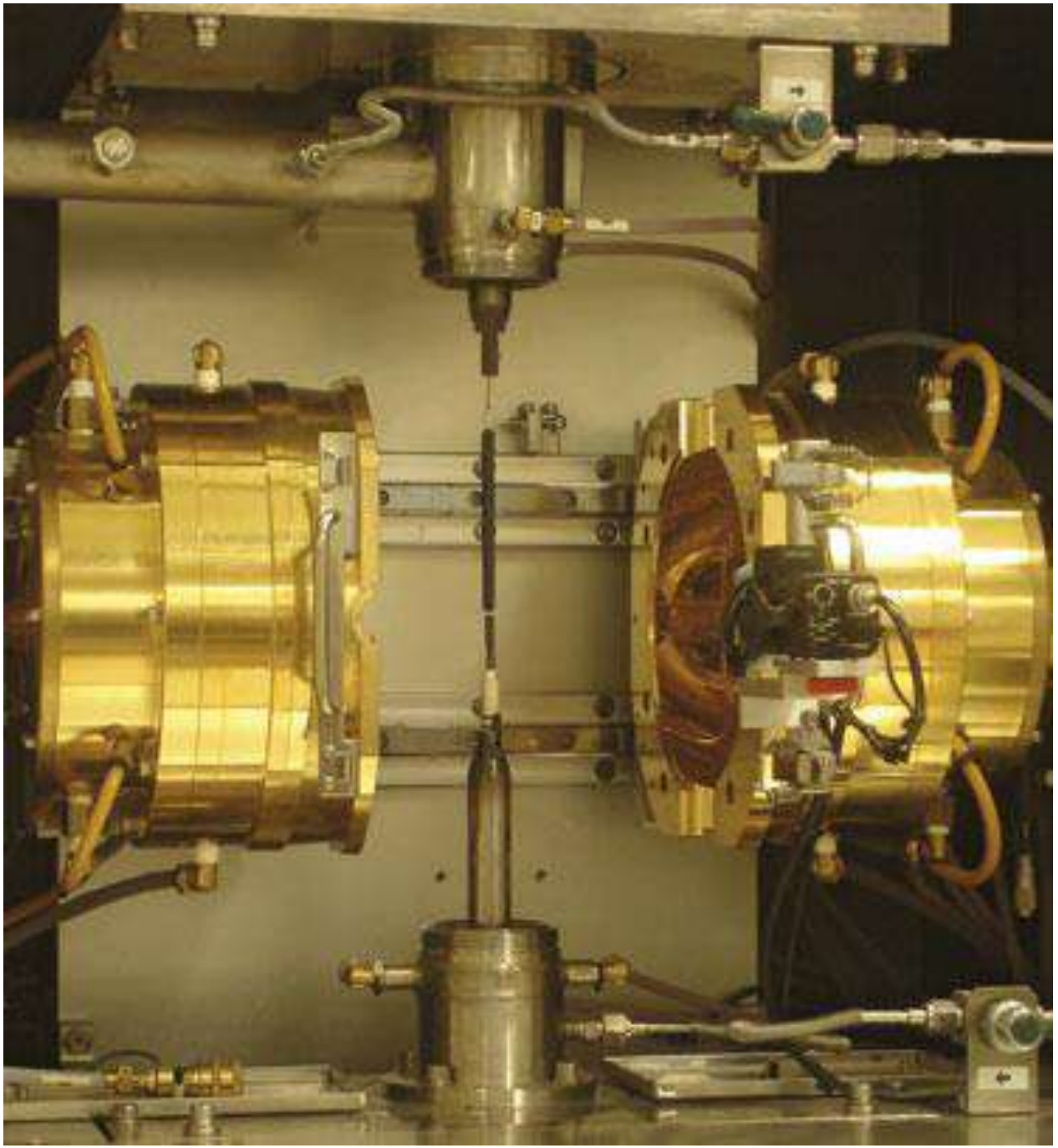




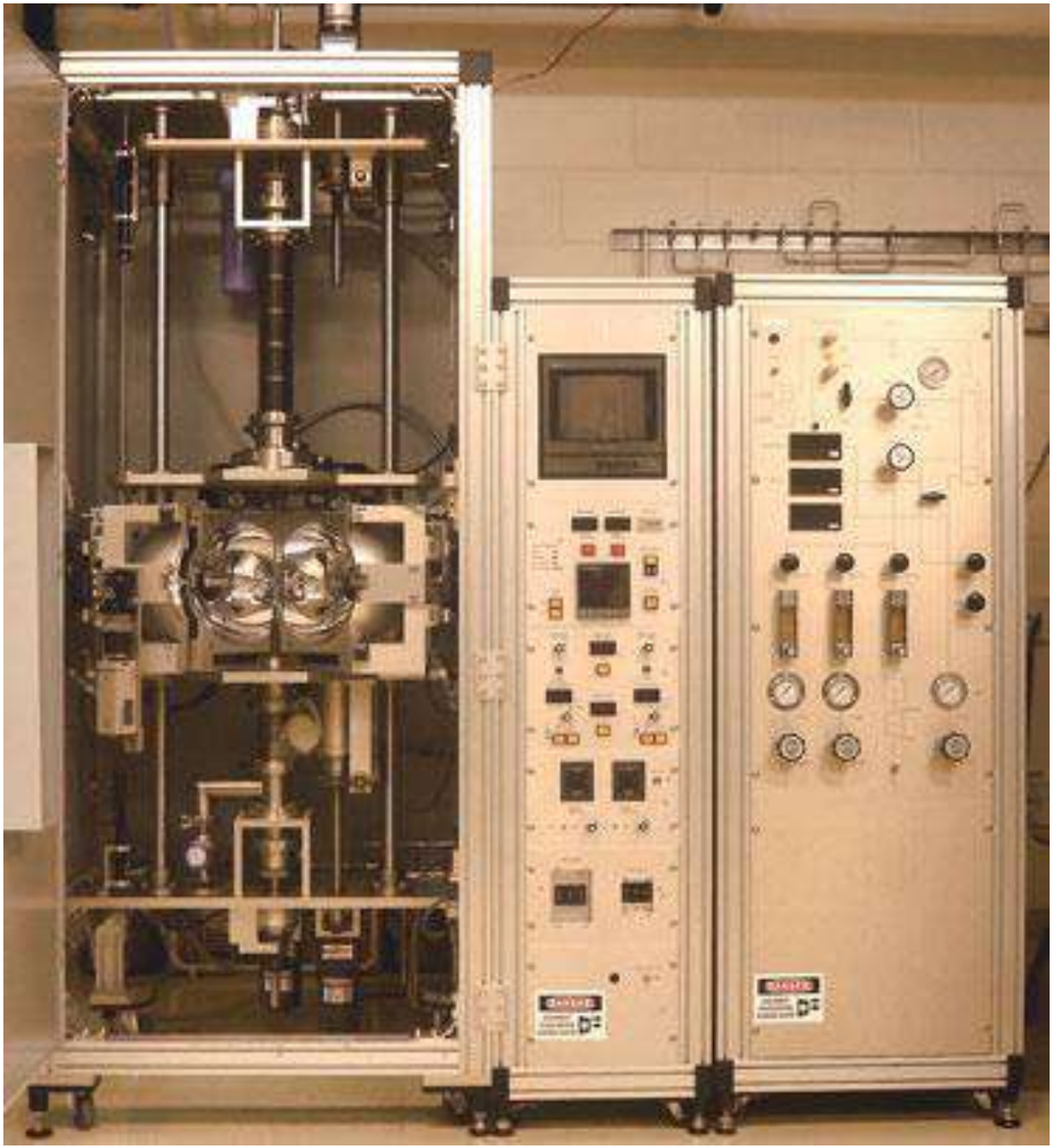


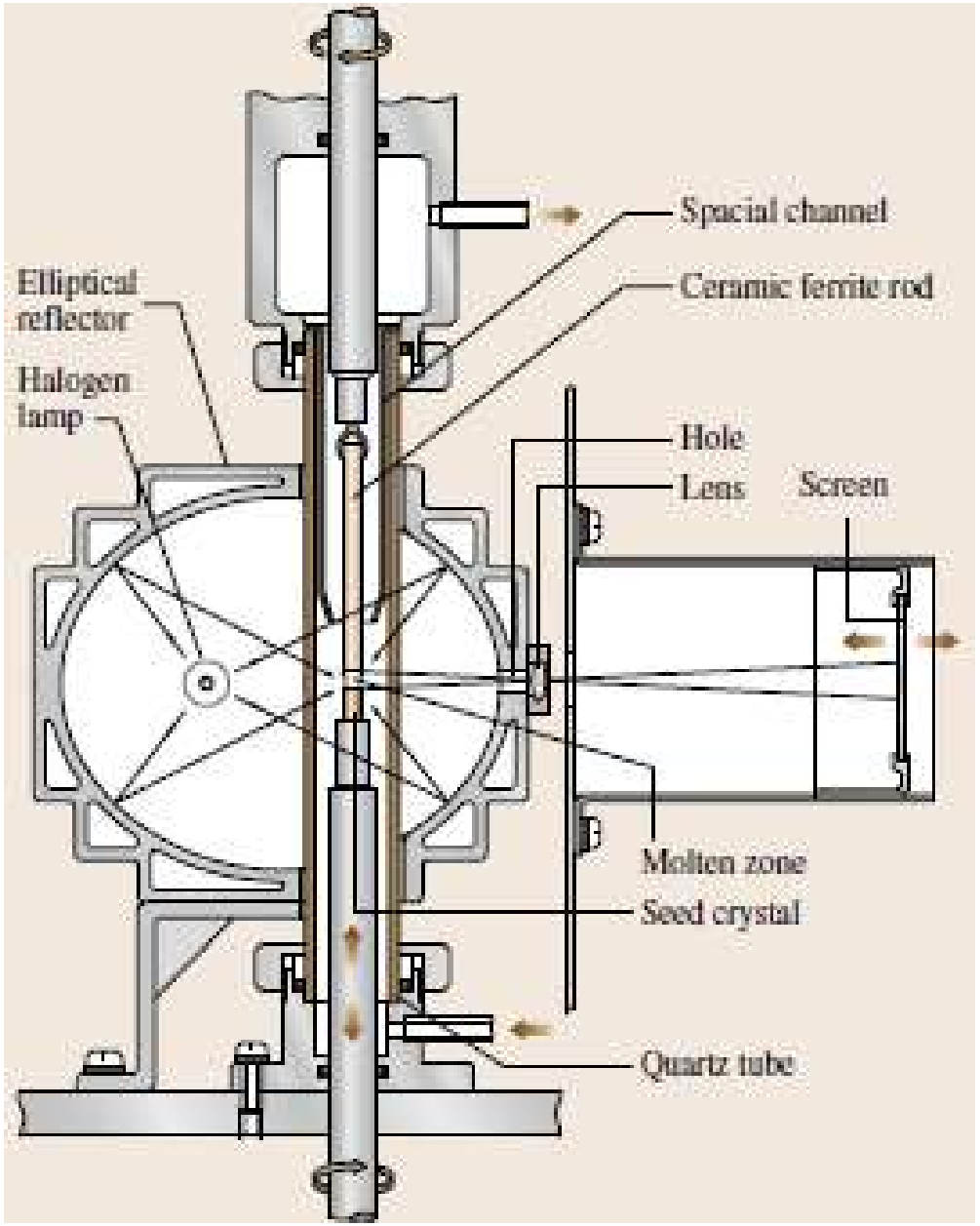


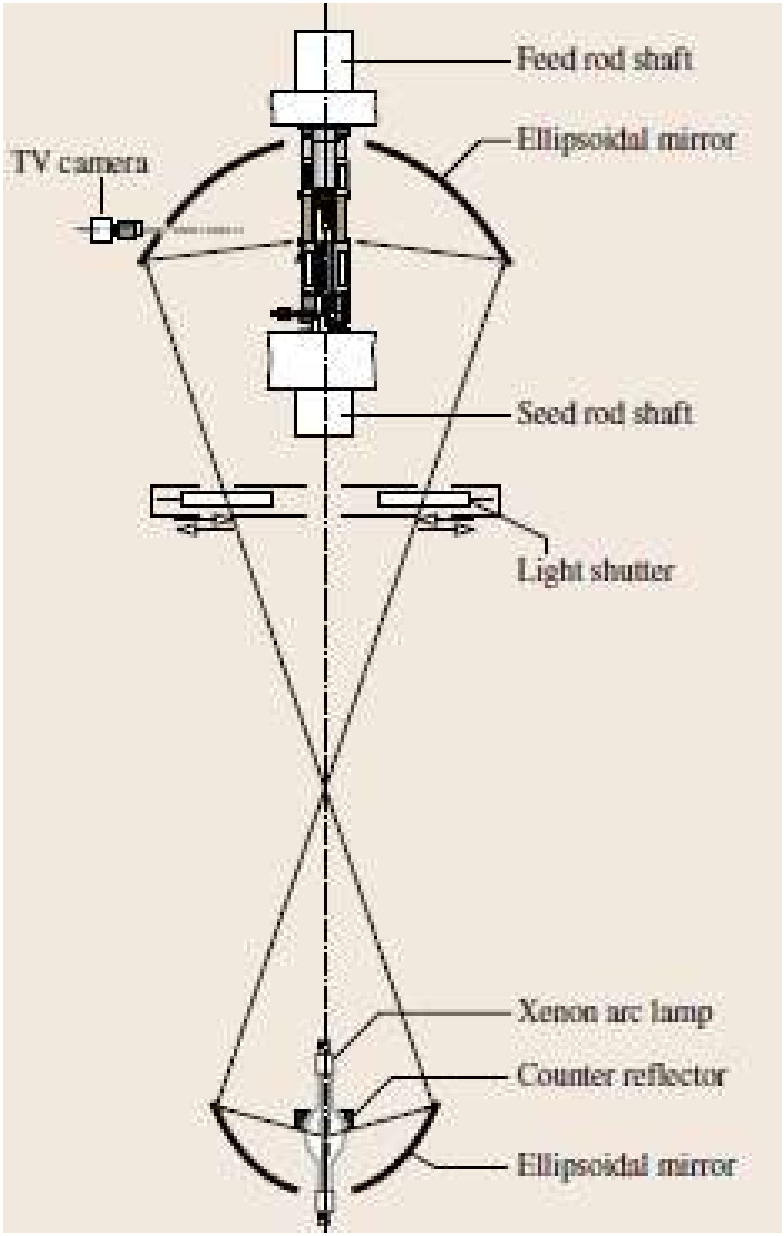

Upper mirror
Pyrometer
Crystal growth chamber
Feed rod
Molten zone
Crystal
Light distribution
Intensity
Mechanical shutter plane
y
x
Pyrometer shutter
z
Lamp
Lower mirror





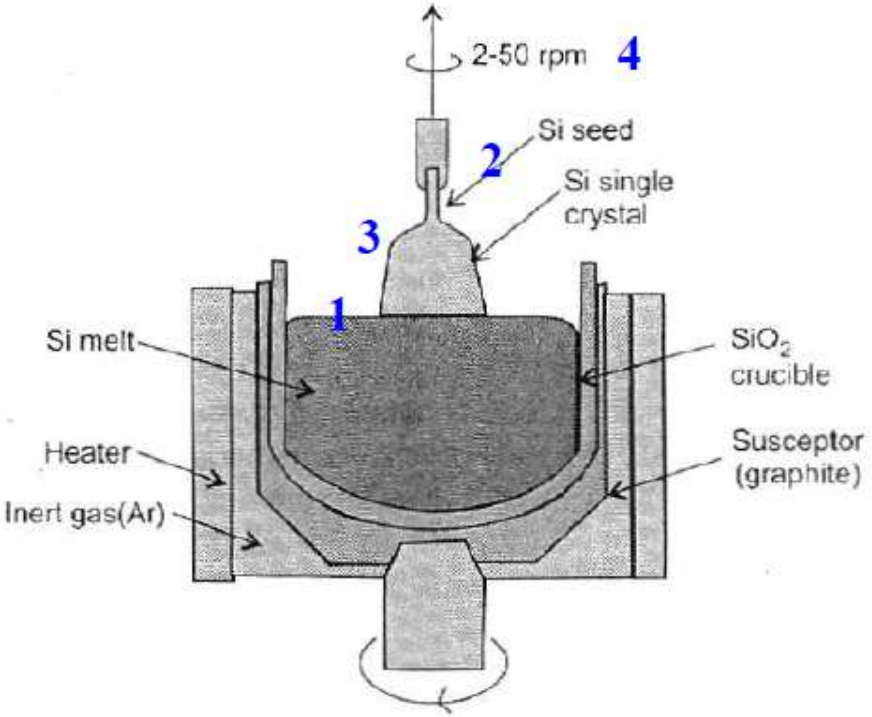



Spacial channel
Ceramic ferrite rod
Elliptical reflector
Halogen lamp
Hole
Lens
Screen
Molten zone
Seed crystal
Quartz tube


Feed rod shaft
Ellipsoidal mirror
TV camera
Seed rod shaft
Light shutter
Xenon arc lamp
Counter reflector
Ellipsoidal mirror

Czochralski

Crystal pulling from congruent melt
Standard method for Si wafers for devices
Melt crystallizes epitaxially on the seed

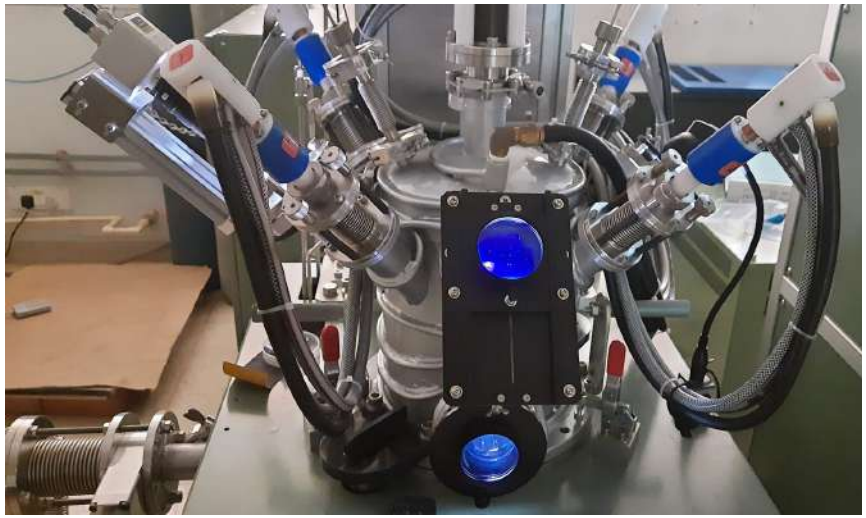


From Yu and Cardona: *Fundamentals of Semiconductors* Springer Verlag 1996

Liquid encapsulated CZ growth: layer of B_2O_3 above the melt helps prevent volatile loss

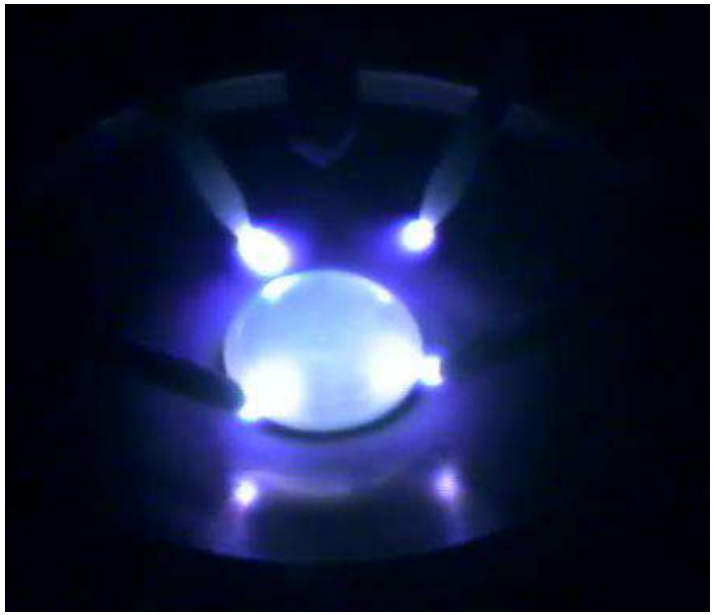
- **1** Cylindrically shaped boules pulled from a bath of molten Si contained in silica crucible under Ar atmoshpere. Crucible is kept at T slightly above melting point.
- **2** Seed (small crystal of Si) is inserted in a bath, then rotated. Once inserted, thermal shock generates dislocations on the crystal
- **3** Dislocations are eliminated by necking – high pull rate is introduced (4-6mm/min) which reduces diameter of the growing crystals to 25 – 30% of the seed size. High pull rates introduces growth velocity greater than velocity of dislocation motion
- **4** Pulling rod and crystal are oppositely rotated few tens rpm's (homogeneity x,T)
- After necking, usual growth rates are 0.1 to 1mm/min. Both growth rates and temperature of the bath are adjusted
- Si crystals is oriented the same way as seed crystal (melt crystallizes epitaxially)
- Axial and radial temperature gradients can introduce defects and dislocations

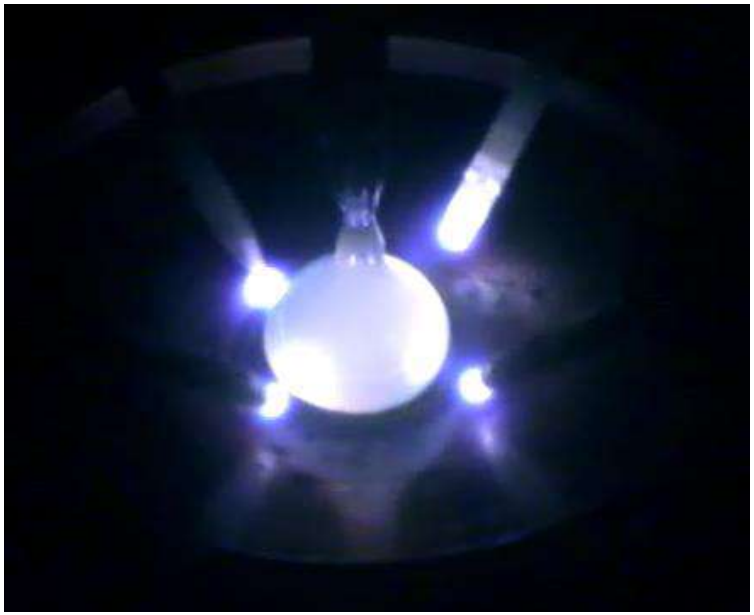

TECHNO SEARCH



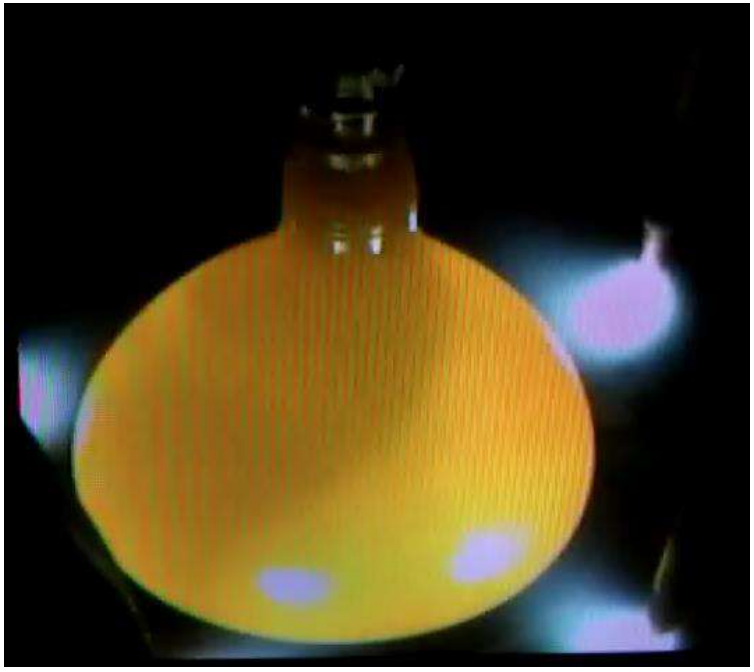




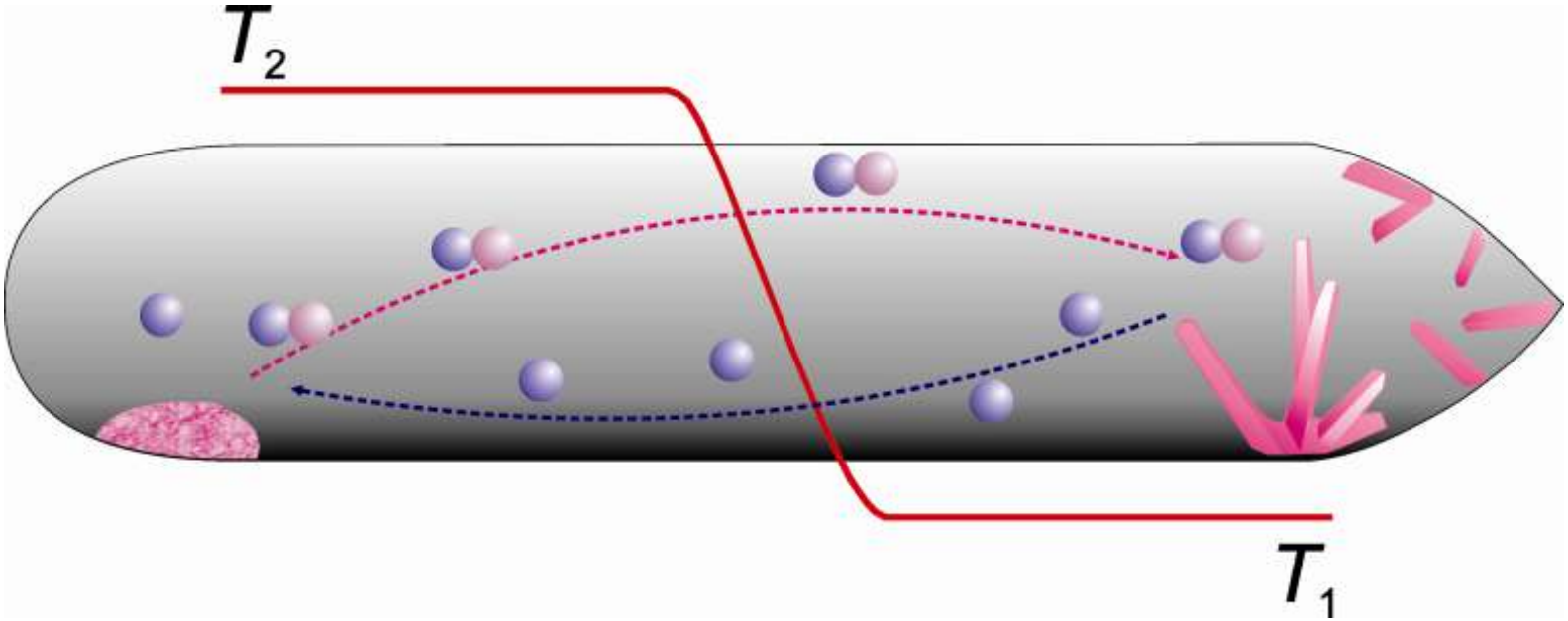








Chemical Vapor Transport



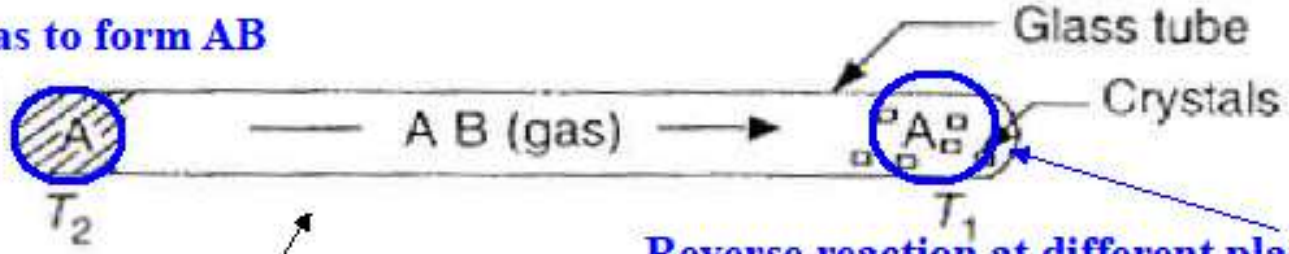


Chemical Vapor Transport Reactions Basics

Phase1: Heterogeneous reaction on the starting solid
Phase 2: Gas motion
Phase 3:Heterogeneous reverse reaction where the solid is formed

Example: Vapor phase transport of a solid substance A via gaseous intermediate AB formed by reaction with vapor transport agent B

A reacts with a gas to form AB



Reverse reaction at different place (sublimation) results in reformation of A

Thermal diffusion:
Consider endothermic reaction:
$Zn\ (l) + S\ (l) \rightarrow ZnS\ (s)$ at 800°C, but faster with I_2 present (**0.5-5mg/cm^3 volume of tube**)

Other transport agents: Br_2, NH_4Cl, $TeCl_3$ or K_2PbCl_6 (for Cl), HCl (using extra capilaries and valves)

Two types of vapor transport reactions:

Endothermic $\Delta T = T_2 - T_1 > 0$
Synthesis of PtO_2: $Pt\ (s) + O_2\ (g) \leftrightarrow PtO_2\ (g)$ (at 1200°C)

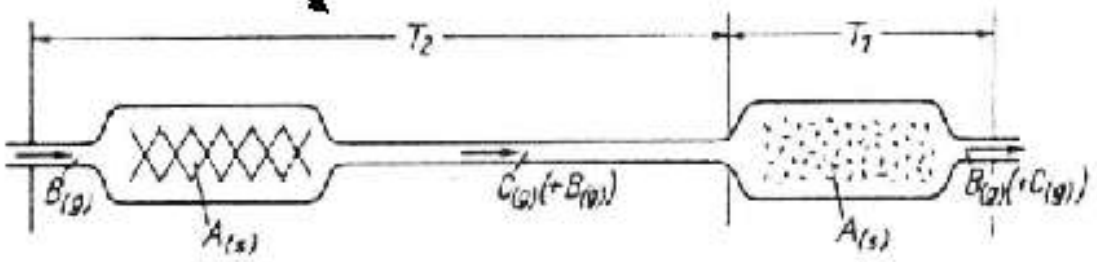
Exothermic $\Delta T = T_2 - T_1 < 0$
Cr purification: $Cr\ (s) + I_2\ (g) \leftrightarrow CrI_2\ (g)$

Chemical Vapor Transport Reactions Basics

Two transport mechanisms of AB:

1. gas stream in open tubes (when the reverse reaction proceeds rapidly)
2. thermal convection in closed tubes (all other cases)

Flow methods:



Gas flows (2-10 cm^3/h) through ΔT, space where gas can stay for a long time is useful for sublimation. Surface exposure to gas flow enhances reaction at T_2 (loose powders) – example PtO_2 reaction.

Gas B can be added at high T or it can be added to A in solid form initially.

Chemical Vapor Transport Reactions Basics

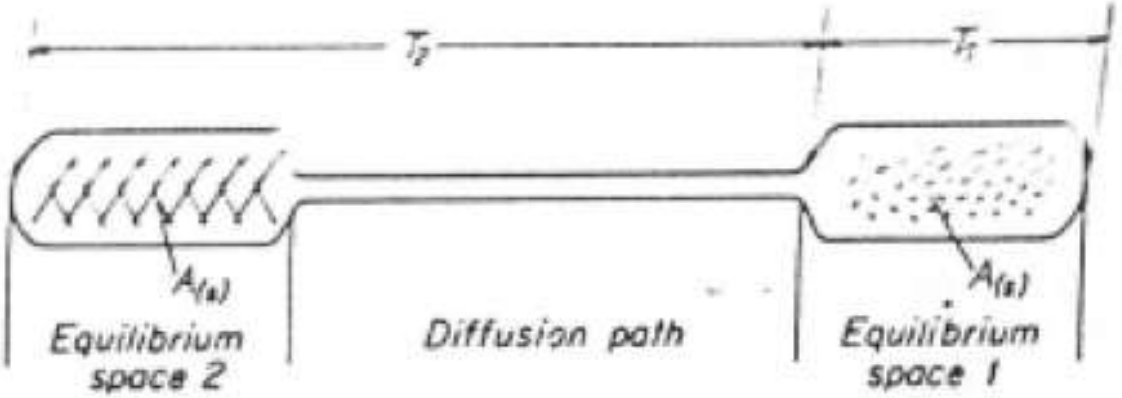
Low P: mean free path of molecules is comparable with tube length – there are no collisions, **material transported is proportional to charge surface area**

Higher P: Gas motion by **diffusion**, for ΔT=const. diffusion rate decreases with P_{total} increase

At even higher P: thermal convection contributes

10^{-3} atm ≤ P ≤ 3 atm.

molecular flow diffusion diffusion + convection

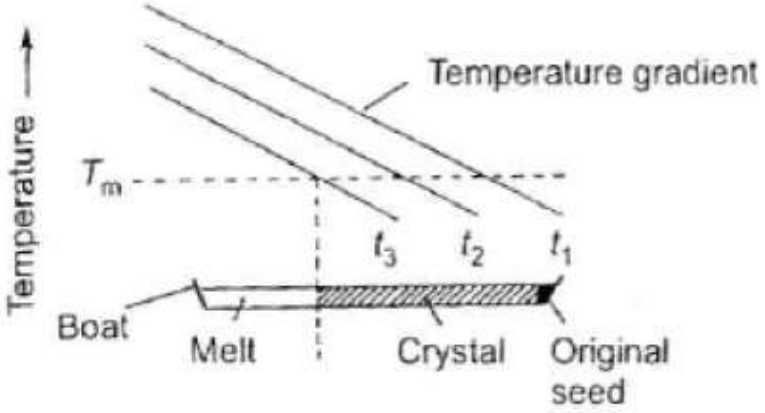


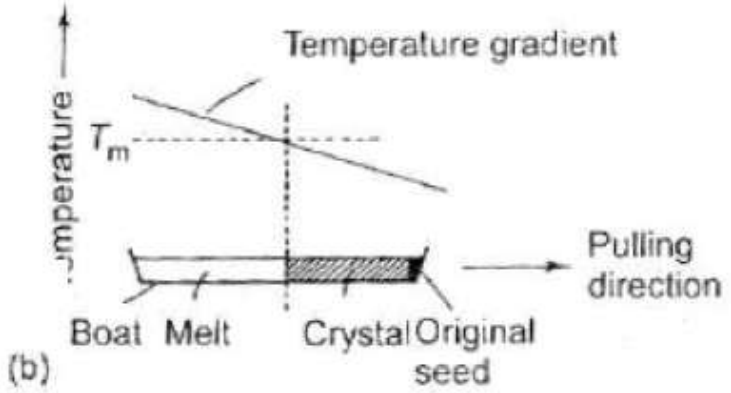
Most applications: closed tubes, P where diffusion dominates

Reactions with (quartz) tube wall need to be considered (eg. Al transport in quartz). Sometimes quartz needs to be protected by alumina, coating.

Bridgeman and Stockbarger

Crystal pulling from congruent melt
Temperature Gradient



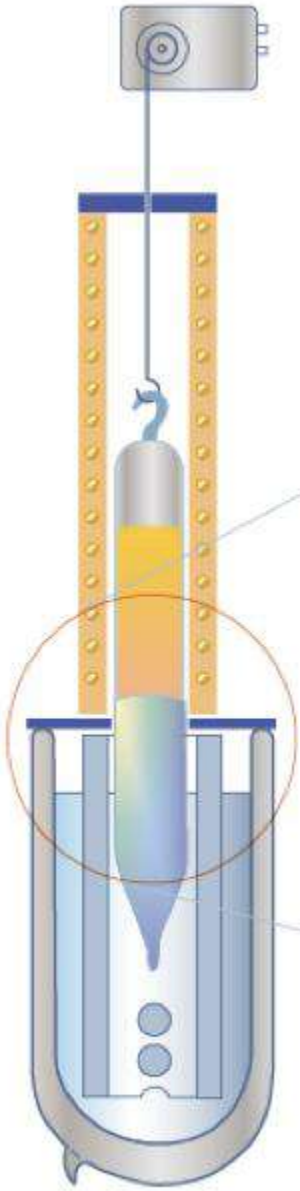


Bridgeman

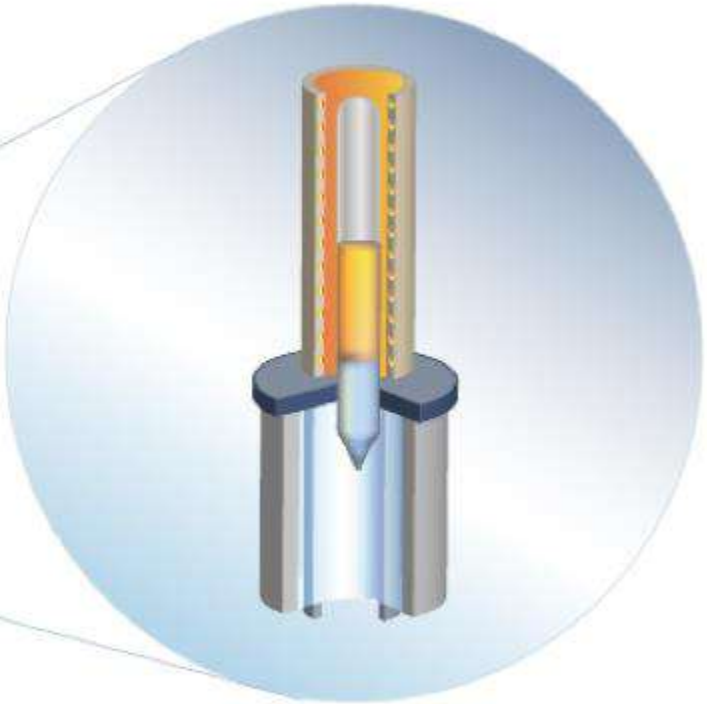
T gradient (ΔT) along crucible is kept constant while average T of the furnace is reduced with time

Crystallization front (melting T) moves progressively along the length of the crucible

Stockbarger method: crucible is moved relative to furnace in which both ΔT and T are constant









Thin Films Deposition

Thin film growth techniques

- evaporation/ MBE
- sputtering
- pulsed laser deposition
- CVD (chemical vapour deposition)

vacuum
- one step epitaxy

- electrochemical
- sol-gel
- spin-coating
- spray pyrolysis

"bulk" techniques
- post-anneal needed for epitaxy

Film growth issues:

1. **Stoichiometry**: chemical composition of the film.
2. **Epitaxy:** the film is single crystal, and is related to the substrate crystal orientation. Achieved by heating the substrate, so the film atoms have energy to relax into a crystal structure commensurate with the substrate.
3. **Impurities:** at high temperatures, the substrate and film atoms can interdiffuse. Other impurity sources: target holders, walls etc.
4. **Strain:** the crystal spacing of film and substrate material is usually not the same. This means the film has intrinsic strain, which can lead to defects, film roughness, delamination.
5. **Second phases:** the film may phase-separate into different regions, with different properties.

1. Island growth (Volmer - Weber)

•form three dimensional islands

•film atoms more strongly bound to each other than to substrate

•and/or slow diffusion



2. Layer by layer growth (Frank - van der Merwe)

•generally highest crystalline quality

•film atoms more strongly bound to substrate than to each other

•and/or fast diffusion



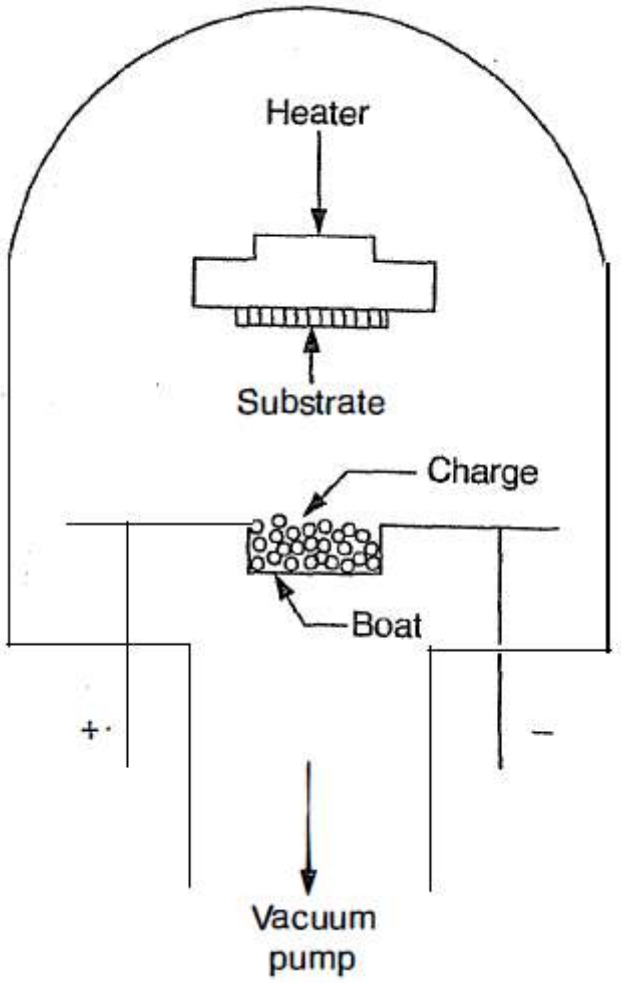
•3. Mixed growth (Stranski - Krastanov)

•initially layer by layer, then forms three dimensional islands

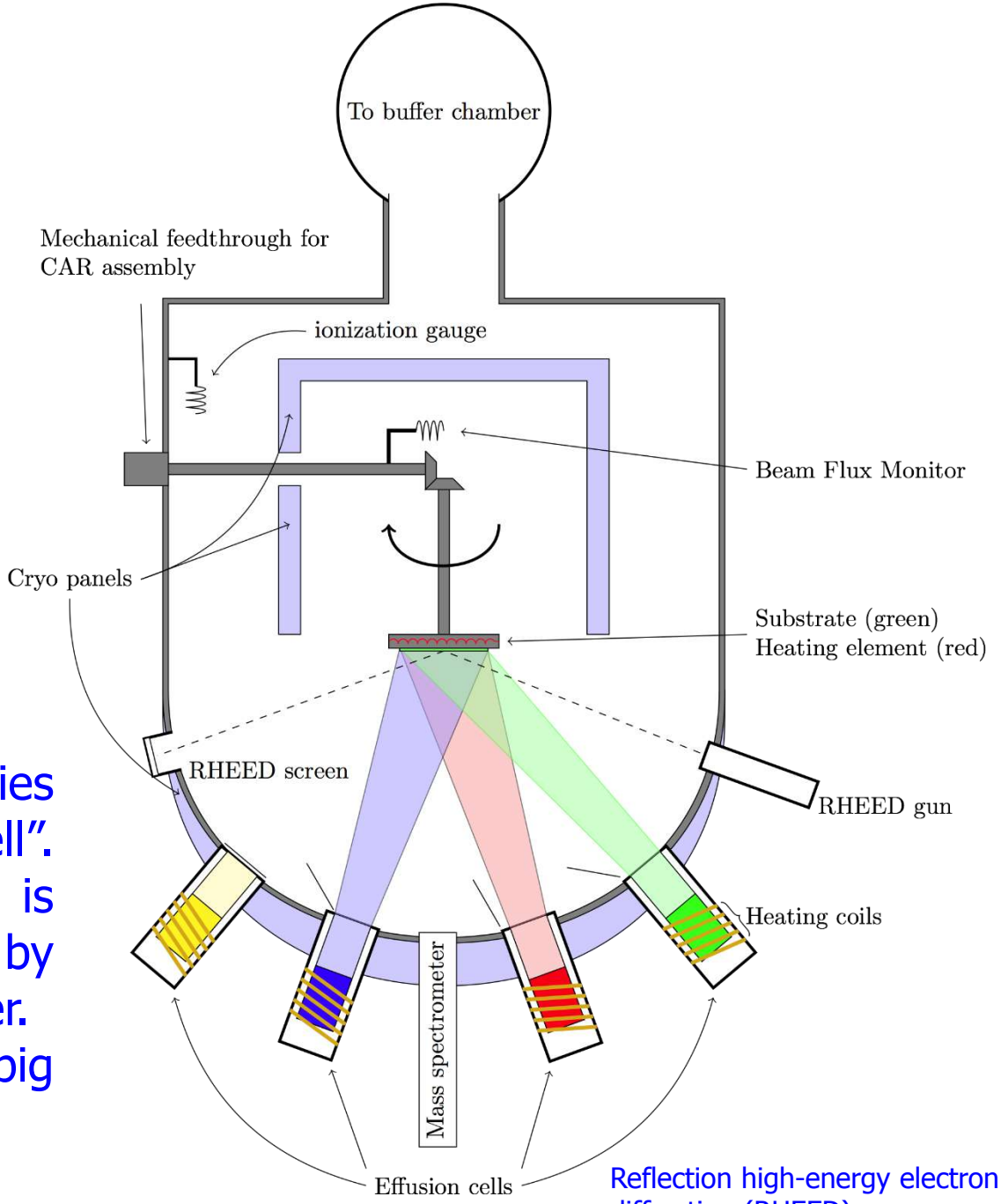


Thermal-evaporation

The starting material, usually in the form of a solid powder.is placed in a 'boat' made from a refractory metal (e.g. Mo or Ta) which is resistively heated in the case of solid charges having relatively low melting points (say~1000 °C). The ensuing vapour, resulting from evaporation of the molten solid contained within the boat, then strikes a substrate positioned over the boat, where it condenses and forms a thin solid film; deposition rate of 0. 1-1 μm/s are typical. In order to achieve compositional homogeneity and thickness uniformity, the substrate is often rotated about an axis normal to the plane of the substrate.



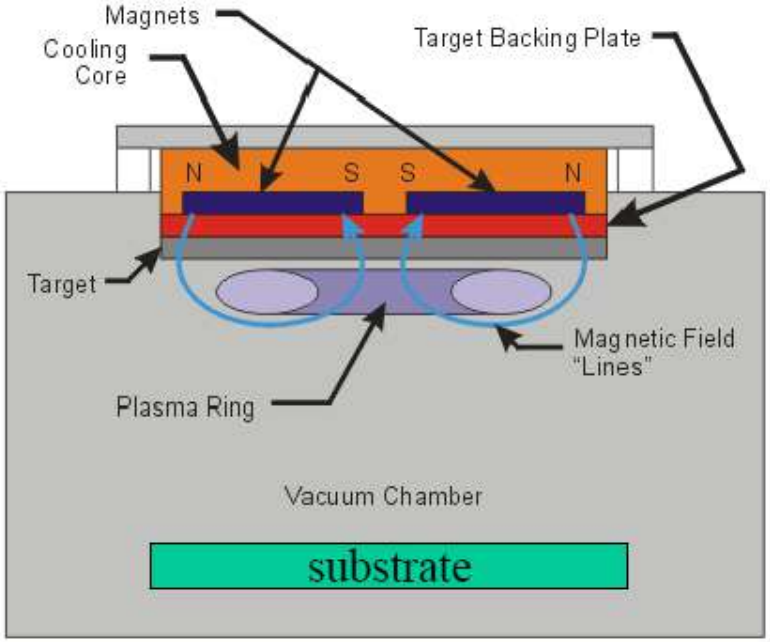
Molecular beam epitaxy

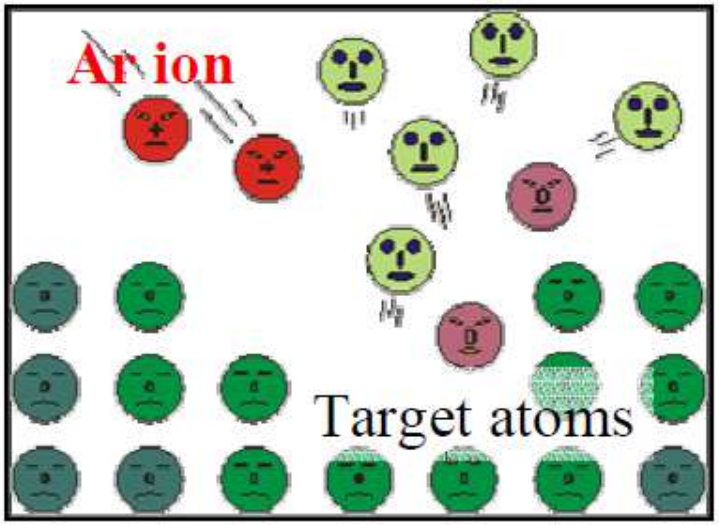


Each target atomic species has its own "effusion cell". Each type of target is vaporized independently by i) heat, ii) e-beam, iii) laser. Rate control is the big challenge.

Reflection high-energy electron diffraction (RHEED)

Sputtering



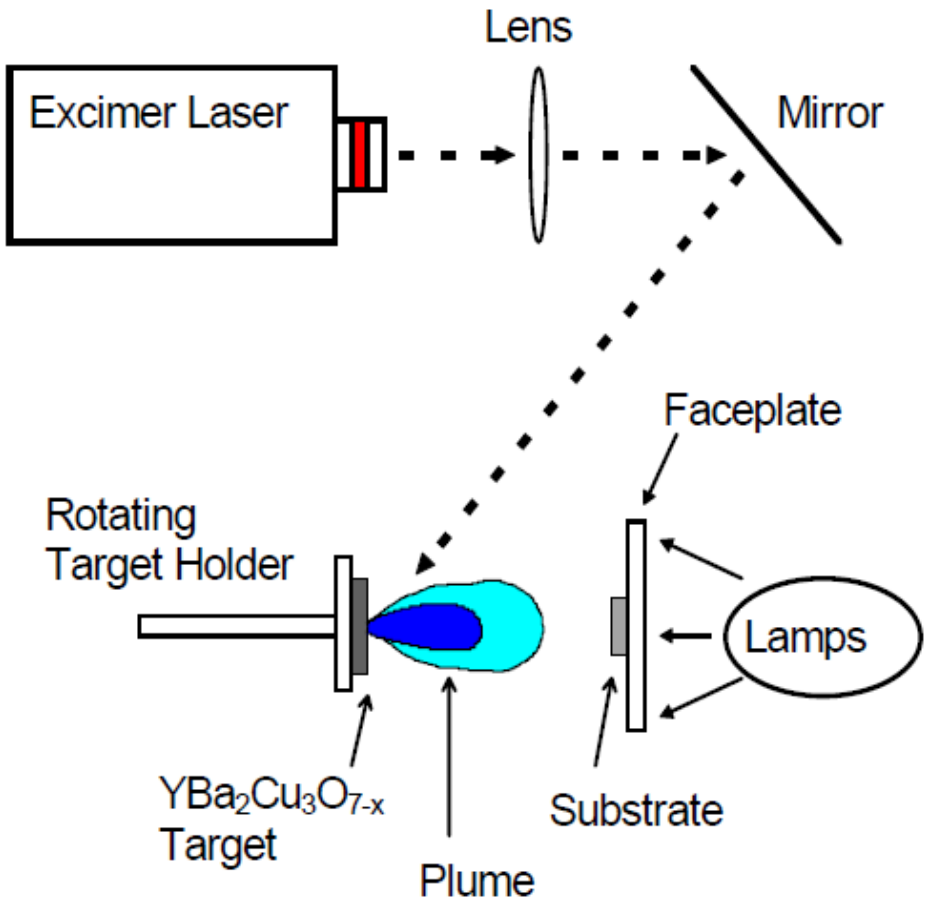


A sputtering gas is used to excite a charged plasma which coats the substrate.

Each type of atom has a different sputtering energy.

Controlling the charged plasma, and preventing resputtering from the substrate are the challenges here.

Pulsed Laser Ablation



A high-power excimer laser is focused on the target.

The target is ablated to form a plume of atoms, molecules and "chunks".

The advantage of PLD is that complex materials can be easily ablated.
The challenge is minimizing "chunks", and maintaining stoichometry.